School of Computing

MIT Vishwaprayag University

Course Plan for Even Semester

Academic Year 2025-26

School of Computing

Bachelor of Computer Applications (BCA)

Message from Registrar

Dear Student,

Welcome back to the your academic endeavors after completing the first semester of AY-2025-26!

We are delighted to help you in your further academic journey through this academic handbook for the “Semester Vasant” of AY-2025-26.

The academic rigor and your learning experiences during Semester Varsha should guide you to use your successes for further aspirations and shortcomings for further improvements.

To help you undertake this academic journey smoothly, I am re-iterating the following points.

- Punctuality, honouring the commitments and self-reporting are the key parameters defining professionalism and you are required to practice these values.
- The academic calendar given to you gives you the schedule of events throughout the semester and you may refer them frequently for your academic preparations and personal commitments.
- Attending all theory and lab sessions is required for your academic progress and performance. Please note as per the regulations of the university and UGC, a minimum of 75% attendance in each course is mandatory for you to be eligible to appear for the end-semester examinations.
- You will be evaluated for your performance in each course through (a) formative assessment during the semester, which is a continuous assessment and (b) summative assessment in the form end-semester evaluation. The details of these assessments for each course can be found in this handbook. You are advised to perform well in all the assessments to get good final grades.
- Strictly avoid falling into detention list due to attendance and/or CIA, which will affect you and your family both emotionally and financially. You are advised to consult the Program Head for understanding the consequences in this regard.
- You must have now noticed that the University is striving to create a conducive atmosphere in the campus and has zero tolerance to misconduct. We believe that peace is a pre-requisite to progress. In view of this, I would like to bring to your attention again the following regulations of the university regarding the code of conduct and discipline and possible consequences of violation.
    a) *Students shall conduct themselves within and outside the premises of the Institute in a manner befitting the students of the University.*

b) *As per the order of Honorable Supreme Court of India, ragging in any form is considered as a criminal offence and is banned. Any form of ragging will be severely dealt with.*

c) *The following acts of omission and/or commission shall constitute gross violation of the code of conduct and are liable to invoke disciplinary measures:*

- *Ragging of any form*
- *Lack of courtesy and decorum; indecent behavior anywhere within or outside the campus.*
- *Willful damage or stealthy removal of any property/belongings of the University/Hostel or of fellow students/citizens.*
- *Possession, consumption or distribution of alcoholic drinks or any kind of narcotics or hallucinogenic drugs.*
- *Mutilation or unauthorized possession of library books.*
- *Noisy and unseemly behavior, disturbing studies of fellow students.*
- *Hacking in computer systems (such as entering into other person's area without prior permission, manipulation and /or damage of computer hardware and software or any other cyber crime etc.)*
- *Plagiarism of any nature.*
- *Any other act of gross indiscipline as decided by the Academic Council from time to time.*

d) *Commensurate with the gravity of offense, the punishment may be: reprimand, fine, expulsion from the hostel, debarring from an examination, disallowing the use of certain facilities of the University, rustication for a specified period or even outright expulsion from the University, or even handing over the case to appropriate law enforcement authorities or the judiciary, as required by the circumstances.*

e) *For an offence committed in (i) a hostel (ii) a School or in a class room and (iii) elsewhere, the Chief Warden, the Program Head/Dean of the School and the Pro VC (Students Wellbeing), respectively, shall have the authority to reprimand or impose fine.*

f) *The University Level Standing Disciplinary Action Committee constituted by the Vice Chancellor, shall be the authority to investigate the details of the offence, and recommend disciplinary action based on the nature and extent of the offence committed.*

Wishing you a continued joyful learning experience in the University!

Registrar

Message from Dean

With the tremendous advancements in computing technology over the past decade, applications of artificial intelligence and machine learning algorithms have proliferated exponentially across literally all walks of life – from healthcare to defence. This has created an unprecedented requirement for experts in this digital computing arena.

We at the School of Computing, MIT Vishwaprayag University, Solapur, are committed to equipping our students with the skills and knowledge needed to thrive in an era of the fast-evolving digital computing landscape. Our programs are designed to help the student advance one's knowledge in topics of one's choice and feature deep insights and accelerated learning formats from renowned academic experts and industry veterans. The structure of the programs at the SoC and the seamless opportunity to collaborate with other Schools at the University, provide naturally, an excellent platform for multi-disciplinary learning – which is very much required in today's world.

Prof. Dr. Ramesh Galigekere
Dean
School of Computing
MIT Vishwaprayag University

Table of Contents

BCA

SEM – IV

S. No.	Course Code	Course Type	Course Title	Credit Scheme of Student Engagement*			
				L	T	P	Total
1	AS2004	BS	Linear Algebra and Calculus	3	0	0	3
2	SC2401	PC	Object Oriented Software Engineering	3	1	0	4
3	SC2113	PC	Advanced Database Concepts	0	0	3	3
4	SC2037	PC	Full Stack Development II	0	0	2	2
5	SC2301	PC	Exploratory Data Analysis and Data Visualization	0	0	2	2
6	SC2303	PE	Introduction to Data Science	3	0	0	3
7		OE	Business Communication	3	0	0	3
				14	**0**	**11**	**24**

Normally one credit refers to one hour instruction for Lecture (L)/Tutorial (T) and two hours instruction for Practical (P) per week

Academic Calendar

CALENDAR OF EVENTS.										
SEMESTER VASANT(EVEN) 2025-26										
UG: B.Tech, BCA, BBA, B.Pharm, B.Des - 2nd/4th/6th Semester										
PG: MCA, MBA, -2nd/4th Semester										
Week No.	**Month**	**Week Days.**							**No Of Working Days**	**Events**
		MON	**TUE**	**WED**	**THU**	**FRI**	**SAT**	**SUN**		
0	Jan/Feb	26	27	28	29	30	31	1		**Jan 26: Research excellence Day, Preparation week**
1	Feb	**2**	3	4	5	6	7	8	5	**Feb 2: Semester opening Day; Feb 2 - 4: Course Registration**
2	Feb	9	10	11	12	13	14	15	6	
3	Feb	16	17	18	19	20	21	22	4	**19th Feb: Chatrapati Shivaji Maharaj jayanti,Feb 16: Last date for withdrawal/change in course Registration**
4	Feb/Mar	23	24	25	26	27	28	1	6	**Mar 28:Thursday time-table**
5	Mar	2	3	4	5	6	7	8	6	**Mar 2- 3: CCM-1,Mar-3: Monthly attendance report, Mar 4: Holi Mar 5-6: Formative Feedback, Mar 7-8: Vishwaprayag Vasantotsav**
6	Mar	9	10	11	12	13	14	15	4	**Mar 14 : Tuesday time table, Mar 9-10: Compensatory holidays**
7	Mar	16	17	18	19	20	21	22	4	**Mar-19 Gudi Padwa**
8	Mar	23	24	25	26	27	28	29	6	**Mar 23-25: Minor-1, Mar 28: Thursday Time table**
9	Mar/Apr	30	31	1	2	3	4	5	4	**Mar 30: Display of marks of Minor -1, Apr-3: Monthly attendance report, March 31: Mahavir Jayanti, April-1: Last date for Make Up Minor-1 registration**
10	Apr	6	7	8	9	10	11	12	6	**Apr-6,7 & 8: Make Up Minor -1, Apr 11: Tuesday Time Table, Apr 6-7: CCM-2**
11	Apr	13	14	15	16	17	18	19	4	**Apr-14: Ambedkar Jayanti**
12	Apr	20	21	22	23	24	25	26	6	**Apr 25: Tuesday Time Table**
13	Apr/May	27	28	29	30	1	2	3	4	**May -1 : Maharashtra Day**
14	May	4	5	6	7	8	9	10	6	**May 4: Monthly attendance report, May 9: Friday time table**
15	May	11	12	13	14	15	16	17	5	**May 11,12 & 13: Minor-2, May 14-15: CCM-3**
16	May	18	19	20	21	22	23	24	6	**May 18: Display of Marks of Minor -2,May 19-20: Summative feedback, May 20: Last date for Make Up Minor-2 Registration , May 21,22,23: Make up Minor 2, May 23: Last date for Course withdrawal, May 23: Display of provisional detention list, May 19-Last date for course withdrawal**
17	May	25	26	27	28	29	30	31	5	**May 27: Bakrid, May 30:Display of final detention list**
18	June	1	2	**3**	4	5	6		3	**June 3 End of Term**
No Of Working Days		**18**	**16**	15	**15**	**15**	**9**		90	

Note: Schools may include school level activities if any, subject to alignment with University-level events and working days.

Important Dates :

COURSES	TERM COMMENCEMENT	END OF TERM	DATE OF COMMENCEMENT	
			THEORY EXAMS	PRACTICAL EXAMS
EVEN SEM: PG: MBA, MCA, **UG:** B.Tech,BCA, BBA	2nd Feb 2026	3rd June, 2026	4th June, 2026	17th June, 2026
SUMMER TERM	22nd June 2026	31st July 2026	3rd August 2026 to 8th Aug 2026	

	HOLIDAYS	MARKS/ATTENDANCE DISPLAY	MINOR EXAM	FEEDBACK	ACADEMIC COMPENSATION

8/1/2026 **Registrar**

SCBTCSE25-26/Rev.1.0

Course Plan

Semester: **IV**	Year: **2026-2027**
Course Title: **Linear Algebra and Calculus**	Course Code: **AS2004**
Total Contact Hours: **60**	Duration of TEE: 3 Hours
TEE Marks: **50**	CIA Marks: **50**
Lesson Plan Author: Dr. Arpita Mandal	Attendance Marks: **0**
Checked By: Mr. Susobhan Pratihar	Last Reviewed Date: 23-01-2026
Co-course Owner:	Last Modified Date: 20-01-2026

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Apply vector and matrix operations, determinants, and eigenvalue methods to solve engineering and physics problems, such as analyzing forces, rotations, and network systems.
2. Use orthogonal vectors and the Gram-Schmidt process to design and optimize signal processing systems, data compression, and feature extraction in machine learning.
3. Employ calculus concepts—limits, derivatives, and integrals—to model and solve real-world problems involving motion, growth, area, and volume calculations in engineering and science.
4. Utilize gradient vectors, Hessian matrices, and second-order derivatives to optimize economic, mechanical, and machine learning systems, including minimizing cost functions and maximizing efficiency.
5. Formulate and solve multivariable optimization problems in practical scenarios, such as resource allocation, predictive modeling, and AI algorithm tuning, integrating linear algebra and calculus techniques.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: **Linear Algebra and Calculus**	Semester: 1 - Semester
Course Code: AS2004	Year: 2025-2026

Course Outcomes (COs) / Program Outcomes (POs)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**PSO1**	**PSO2**
1. Apply vector and matrix operations, determinants, and eigenvalue methods to solve engineering and physics problems, such as analyzing forces, rotations, and network systems.	H				M									
2. Use orthogonal vectors and the Gram-Schmidt process to design and optimize signal processing systems, data compression, and feature	H				M									

extraction in machine learning.														
3. Employ calculus concepts—limits, derivatives, and integrals—to model and solve real-world problems involving motion, growth, area, and volume calculations in engineering and science.	H				M									
4. Utilize gradient vectors, Hessian matrices, and second-order derivatives to optimize economic, mechanical, and machine learning systems, including minimizing cost functions and maximizing efficiency.	H				M									
5. Formulate and solve multivariable	H				M									

optimization problems in practical scenarios, such as resource allocation, predictive modeling, and AI algorithm tuning, integrating linear algebra and calculus techniques.														

Course Content

Course Code: AS2004		Course Title: **Linear Algebra and Calculus**	
L-T-P: 3-1-0		Credits: 4	Contact Hrs: 60
CIA Marks: 50	TEE Marks: 50	Attendance Marks: 0	Total Marks: 100
Teaching Hrs: 60		Exam Duration: 3 hrs	

Content	Hrs
Unit I- **Linear Algebra Foundations**	
Chapter No. 1 - Vectors and Matrices 1. **Vectors:** Definition, representation in 2D, 3D, and n-dimensions; types (zero, unit, position, equal, negative); operations (addition, subtraction, scalar multiplication) and their properties. 2. **Vector Products:** Dot product and cross product with properties and applications. 3. **Matrices:** Definition, notation, and types (row, column, square, diagonal, scalar, identity, zero, symmetric, skew-symmetric, triangular); matrix operations (addition, subtraction, scalar multiplication, matrix multiplication) and their properties. 4. **Determinants and Inverse:** Determinants, inverse of a matrix, conditions for invertibility, and basic properties. 5. **Linear Systems and Row Reduction:** Solving linear systems (Ax = b) using Gaussian elimination, Cramer's rule, and inverse method; elementary row operations; row-echelon and reduced row-echelon forms; rank and its significance.	12 hrs
Chapter No. 2 - Eigenvalues, Eigenvectors, and Orthogonalization 1. **Eigenvalues and Eigenvectors:** Definition of eigenvalues and eigenvectors of a square matrix; characteristic equation; computation methods for small matrices. 2. **Properties of Eigenvalues and Eigenvectors:** Sum and product of eigenvalues, linear independence of eigenvectors corresponding to distinct eigenvalues, diagonalizability conditions. 3. **Diagonalization:** Definition of diagonalizable matrices; conditions for diagonalization; process of diagonalization using eigenvalues and	12 hrs

eigenvectors; applications in simplifying matrix powers and linear transformations. 4. **Orthogonal Vectors and Orthogonal Matrices:** Definition of orthogonal and orthonormal vectors; properties of orthogonal matrices; applications in simplifying computations and preserving vector lengths. 5. **Gram-Schmidt Process:** Procedure to convert a set of linearly independent vectors into an orthonormal set; applications in QR decomposition, projections, and numerical computations	
Unit II- **Introduction to Calculus**	
Chapter No. 3 - Functions, Calculus, and Applications 1. **Functions, Domain, and Range:** Definition of functions; types (one-to-one, onto, bijective); domain, range, and inverse functions; composition of functions. 2. **Limits and Continuity:** Concept of limit of a function; techniques for evaluating limits; continuity of a function; types of discontinuities. 3. **Derivatives and Differentiation Rules:** Definition of derivative; rules of differentiation (product, quotient, chain); higher-order derivatives; partial derivatives. 4. **Applications of Derivatives:** Use of derivatives in optimization problems (finding maxima and minima); tangent and normal lines; motion along a line; related rates. 5. **Double and Triple Integrals:** Evaluation of double and triple integrals; applications in computing areas, volumes, and physical quantities; iterated integrals and change of order of integration.	12 hrs
Chapter No. 4 - Gradients, Hessians, and Optimization 1. **Gradient Vectors:** Definition of gradient of a scalar function; geometric interpretation; direction of maximum increase; significance of gradients in optimization problems. 2. **Directional Derivatives:** Definition and computation of directional derivatives; relationship with gradient vectors; applications in multivariable function analysis. 3. **Second-Order Derivatives:** Second-order partial derivatives; mixed derivatives; Clairaut's theorem; Taylor expansion of multivariable functions.	12 hrs
Unit III- **Calculus for Machine Learning**	

Chapter No. 5 - Optimization and Machine Learning Applications 1. **Introduction to Optimization Problems:** Definition and formulation of optimization problems; objective functions; constraints; classification of optimization problems. 2. **Unconstrained and Constrained Optimization:** Optimization of single and multivariable functions; critical points; basic constrained optimization concepts. 3. **Cost Functions in Machine Learning:** Definition of cost (loss) functions; examples from regression and classification; role of cost functions in learning models. 4. **Minimization Techniques:** Gradient-based minimization methods; interpretation of optimization results; applications in training machine learning models.	12 hrs

Chapterwise Plan

Course Code and Title: AS2004 / **Linear Algebra and Calculus**	
Chapter Number and Title: **1 - Vectors and Matrices**	Planned Hours: **12 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Sl. No.	Topic Learning Outcomes	COs	BL
1	Define, classify, and apply vector operations in 2D, 3D, and n-dimensional spaces, and verify their algebraic properties.	CO1	L3
2	Apply dot and cross products to solve engineering problems and identify and classify different types of matrices.	CO1	L3
3	Perform matrix operations and compute determinants to analyze matrix properties and determine invertibility.	CO1	L3
4	Analyse systems of linear equations using matrix methods and apply row reduction techniques to determine echelon forms and matrix rank.	CO1	L4

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to vectors, representation in 2D & 3D
2. n-dimensional vectors, types of vectors
3. Vector operations (addition, subtraction, scalar multiplication)
4. Dot product: definition, properties, applications
5. Cross product: properties and applications
6. Introduction to matrices and classification
7. Matrix operations and properties
8. Inverse of a matrix, conditions for invertibility
9. Determinants: definition and properties

10. Linear systems (Ax = b), Gaussian elimination
11. Row echelon & reduced row echelon forms
12. Rank of a matrix and applications

Review Questions

Sl. No.	- Questions	TLOs	BL
1.	A particle moves from point $P(1,2,3)$to $Q(4,6,8)$. Represent the displacement as a vector and find its magnitude.	TLO1	L3
2	Given vectors $\mathrm{A} = 2\mathrm{i} + 3\mathrm{j} - \mathrm{k}$and $\mathrm{B} = -\mathrm{i} + 4\mathrm{j} + 2\mathrm{k}$, compute $\mathrm{A} + \mathrm{B}$ and $\mathrm{A} - \mathrm{B}$.	TLO1	L3
3.	Find the dot product and cross product of $\mathrm{A} = \mathrm{i} + 2\mathrm{j}$and $\mathrm{B} = 3\mathrm{i} - \mathrm{j}$.	TLO2	L3
4.	Classify the matrices: $\mathrm{M}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \mathrm{M}_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \mathrm{M}_3 = \begin{bmatrix} 2 & 3 \\ 2 & 3 \end{bmatrix}$.	TLO2	L2
5.	Compute AB for $\mathrm{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \mathrm{B} = \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix}$and verify whether $\mathrm{AB} = \mathrm{BA}$.	TLO3	L3
6.	Find the determinant and inverse (if it exists) of $\mathrm{A} = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$.	TLO3	L3
7.	Solve the system of equations using Gaussian elimination: $x + y + z = 6, 2x - y + 3z = 14, -x + 4y + z = -2$.	TLO4	L3
8.	Reduce the matrix $\mathrm{A} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{bmatrix}$to row-echelon form and determine its rank.	TLO4	L4

Course Code and Title: AS2004 / **Linear Algebra and Calculus**	
Chapter Number and Title: **2 - Eigenvalues, Eigenvectors, and Orthogonalization**	Planned Hours: **12 hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Sl. No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Define eigenvalues and eigenvectors of a matrix and compute them for 2×2 and 3×3 matrices using the characteristic equation.	CO1	L2
2	verify properties of eigenvalues and eigenvectors and diagonalize a matrix, where possible, using eigenvalue–eigenvector methods.	CO1	L2
3	Identify orthogonal and orthonormal vector sets and apply the Gram–Schmidt process to orthogonalize linearly independent vectors.	CO2	L3
4	Apply diagonalization and orthogonalization techniques to simplify matrix computations and analyze their practical applications.	CO2	L3

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to eigenvalues and eigenvectors
2. Characteristic equation and eigenvalue computation (2×2)
3. Eigenvalues and eigenvectors of 3×3 matrices
4. Properties of eigenvalues (sum, product)
5. Linear independence of eigenvectors, diagonalizability conditions
6. Diagonalization of matrices
7. Applications of diagonalization (matrix powers, transformations)

8. Orthogonal and orthonormal vectors
9. Orthogonal matrices and their properties
10. Gram–Schmidt orthogonalization process
11. Applications of Gram–Schmidt (QR, projections)
12. Integrated problems using diagonalization & orthogonalization

Review Questions

Sl. No.	Questions	TLOs	BL
1.	Find the eigenvalues and eigenvectors of the matrix $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.	TLO1	L2
2	Determine the eigenvalues and corresponding eigenvectors of $\mathrm{B} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$	TLO1	L1
3.	Verify that the eigenvectors of $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ corresponding to distinct eigenvalues are linearly independent.	TLO2	L3
4.	Diagonalize the matrix $\mathbf{C} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$ using its eigenvalues and eigenvectors.	TLO2	L3
5.	Check if the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ are orthogonal and/or orthonormal.	TLO3	L2
6.	Apply the Gram-Schmidt process to the vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ to obtain an orthonormal set.	TLO3	L3

7.	Use diagonalization to compute $\mathbf{A}^5$ for $\mathbf{A} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$.	TLO4	L3
8.	Analyze the orthogonalization of vectors $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ and discuss how Gram-Schmidt preserves linear independence.	TLO4	L4

Course Code and Title: AS2004 / **Linear Algebra and Calculus**	
Chapter Number and Title: **3 - Functions, Calculus, and Applications**	Planned Hours: **12 hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Sl. No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Define functions and determine their domain, range, and type, and compute limits to analyze continuity at given points.	CO3	L2
2	Apply rules of differentiation to compute derivatives and solve optimization problems involving maxima and minima.	CO3	L3
3	Compute first, higher-order, and partial derivatives, and evaluate double and triple integrals over specified regions.	CO3	L3
4	Apply derivatives and multiple integrals to analyze and solve real-world problems involving area, volume, and optimization.	CO3	L3

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to functions; domain and range
2. Types of functions; inverse and composition
3. Limits of functions and evaluation techniques
4. Continuity and types of discontinuities
5. Definition of derivative; basic differentiation rules
6. Product, quotient, and chain rules
7. Applications of derivatives: maxima and minima
8. Tangent, normal, motion along a line, related rates
9. Higher-order and partial derivatives
10. Double integrals and applications
11. Triple integrals; change of order
12. Applications of derivatives and integrals

Review Questions

	Questions	**TLOs**	**BL**
1.	Determine the domain and range of the function $f(x) = \sqrt{4 - x^2}$ and determine whether it is continuous locally or globally.	TLO1	L2
2.	Compute $\lim_{x \to 0} \frac{\sin 3x}{x}$ and check whether the function $f(x) = \frac{\sin 3x}{x}$ is continuous at $x = 0$.	TLO1	L3
3.	Find the derivative of $f(x) = x^2 \sin x + \frac{\ln x}{x}$ using product, quotient, and chain rules.	TLO2	L3
4.	A farmer wants to maximize the area of a rectangular field adjacent to a river with 100 m of fencing (one side along the river doesn't require fencing). Find the dimensions of the field for maximum area.	TLO2	L3

5.	Compute the first and second derivatives of $f(x) = x^3 e^x$.	TLO3	L3
6.	Evaluate the double integral $\iint_R (x+y)\, dxdy$, where R is the rectangle defined by $0 \le x \le 2, 0 \le y \le 3$.	TLO3	L3
7.	Compute the volume of the solid bounded by $z = x^2 + y^2$and the plane $z = 4$using a triple integral.	TLO4	L3
8.	Analyze the function $f(x) = x^3 - 3x^2 + 4$ to find its critical points and use derivatives to identify maxima and minima.	TLO4	L4

Course Code and Title: AS2004 / **Linear Algebra and Calculus**	
Chapter Number and Title: **4 - Gradients, Hessians, and Optimization**	Planned Hours: **12 hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Define gradient vectors and compute gradients of scalar functions of two or three variables, explaining their geometric and optimization significance.	CO4	L2
2	Determine directional derivatives using gradient vectors and explain the concept and role of the Hessian matrix in multivariable analysis.	CO4	L3
3	Compute the Hessian matrix and analyze critical points using gradient and Hessian methods to classify maxima, minima, and saddle points.	CO4	L3
4	Analyse gradient and Hessian techniques to solve and interpret real-world optimization problems in engineering, economics, and machine learning.	CO4	L4

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to multivariable functions and gradients
2. Computation of gradient vectors (2 & 3 variables)
3. Geometric interpretation of gradients
4. Directional derivatives: definition and computation
5. Relationship between gradient and directional derivative
6. Second-order partial derivatives and mixed derivatives
7. Hessian matrix: definition and computation
8. Critical points and second derivative test
9. Classification of maxima, minima, and saddle points
10. Optimization problems in two variables
11. Applications in economics, mechanics, and ML
12. Integrated problems and recap

Review Questions

Questions		**TLOs**	**BL**
1.	Define the gradient vector for a function $f(x,y) = x^2y + 3y^2$and explain its significance in optimization.	TLO1	L2
2.	Compute the gradient vector $\nabla f(x,y,z)$for $f(x,y,z) = x^2 + y^2 + z^2 - xy + 2z$.	TLO1	L3
3.	Find the directional derivative of $f(x,y) = x^2y + y^2$at point (1,2) in the direction of the vector $\mathrm{v} = 3\mathrm{i} + 4\mathrm{j}$.	TLO2	L3
4.	Define the Hessian matrix for a function $f(x,y) = x^3 + y^3 - 3xy$and explain its role in classifying critical points.	TLO2	L2
5.	Compute the Hessian matrix of $f(x,y) = x^2y + y^3 - x$and evaluate it at the point (1,1).	TLO3	L3

6.	Determine the critical points of $f(x,y) = x^3 - 3xy^2$and classify them as maxima, minima, or saddle points using the gradient and Hessian.	TLO3	L3
7.	Solve an optimization problem: Maximize $f(x,y) = xy - x^2 - y^2$using gradient and Hessian analysis.	TLO4	L3
8.	Discuss how the Hessian matrix and second-order derivatives are used to analyze the stability of a function $f(x,y) = x^2 + xy + y^2$in real-world optimization scenarios.	TLO4	L4

Course Code and Title: AS2004 / **Linear Algebra and Calculus**	
Chapter Number and Title: **5 - Optimization and Machine Learning Applications**	Planned Hours: **12 hrs**

Learning Outcomes: -

At the end of the topic the student should be able to:

Sl. No.	Topic Learning Outcomes	COs	BL
1	Define optimization problems and formulate mathematical models for simple unconstrained and constrained optimization scenarios.	CO5	L2
2	Apply first- and second-order optimality conditions using gradient and Hessian methods to identify and classify critical points.	CO5	L3
3	Define machine learning cost functions and compute their gradients to apply gradient-based minimization techniques.	CO5	L3

4	Solve machine learning optimization problems using gradient-based methods and analyze the convergence and efficiency of optimization algorithms.	CO5	L4

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to optimization problems
2. Objective functions, constraints, and classification
3. Unconstrained optimization and critical points
4. First-order optimality condition (∇f = 0)
5. Second-order conditions and Hessian-based tests
6. Constrained optimization: basic concepts
7. Cost functions in machine learning
8. Gradient computation of cost functions
9. Gradient descent algorithm
10. Applications of gradient descent in ML models
11. Convergence, learning rate, and efficiency issues
12. Integrated ML optimization problems and recap

Review Questions

	Questions	**TLOs**	**BL**
1.	Define local and global maxima and minima for the function $f(x) = x^3 - 6x^2 + 9x$, and identify them.	TLO1	L2
2.	Formulate an optimization problem to maximize the area of a rectangular plot with a fixed perimeter of 100 m.	TLO1	L3
3.	Find the critical points of $f(x, y) = x^2 + xy + y^2 - 4x - 6y$by setting the gradient equal to zero.	TLO2	L3

4.	Use the Hessian matrix to classify the critical points of $f(x, y) = x^3 - 3xy^2$as maxima, minima, or saddle points.	TLO3	L4
5.	Define a cost function $J(\theta) = (\theta - 3)^2$for a simple machine learning problem and explain its significance in optimization.	TLO3	L2
6.	Apply gradient descent to minimize $J(\theta) = (\theta - 3)^2$with initial guess $\theta_0 = 0$and learning rate $\alpha = 0.1$. Compute the first three iterations.	TLO3	L3
7.	Solve a linear regression optimization problem: Minimize $J(\theta_0, \theta_1) = \sum_{i=1}^{n}(y_i - (\theta_0 + \theta_1 x_i))^2$for the dataset $(x_1, y_1) = (1,2), (x_2, y_2) = (2,3)$.	TLO4	L3
8.	Analyze the effect of different learning rates ($\alpha = 0.01, 0.1, 0.5$) on the convergence of gradient descent for $J(\theta) = (\theta - 5)^2$.	TLO4	L4

Model Question Paper

Question Paper Title: MINOR I SEM IV		
Total Duration (H:M): 1:30	Course: **Linear Algebra and Calculus** (AS2004)	Maximum Marks: 40
Note: Attempt ANY TWO questions from Q.1, Q.2 and Q.3		

Q. No.		**Questions**	**Marks**	**CO**	**BL**
1.	1A1.	The zero vector in $\mathbb{R}^3$is a) (1,0,0) b) (0,1,0) c) (0,0,0) d) (1,1,1)	1.00	CO1	L1
	1A2.	A matrix whose determinant is zero is called a) Identity b) Orthogonal c) Singular d) Diagonal	1.00	CO1	L1
	1A3.	Eigenvalues of a diagonal matrix are a) Row sums b) Column sums c) Diagonal elements d) Determinant	1.00	CO2	L2
	1A4.	Two vectors are orthogonal if their dot product is a) 1 b) –1 c) 0 d) ∞	1.00	CO2	L1
	1A5.	Which process converts a set of vectors into an orthogonal set? a) Diagonalization	1.00	CO2	L1

		b) Row reduction c) Gram–Schmidt d) LU decomposition			
	1B.	Let $A = \begin{bmatrix} 4 & 5 \\ 2 & 3 \end{bmatrix}$. Compute Det (A) and determine whether A is invertible.	3.00	CO1	L2
	1C.	Reduce the matrix $M = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{bmatrix}$ to row-echelon form and determine its rank.	5.00	CO2	L3
	1D.	Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.	7.00	CO2	L3
2.	2A1.	The rank of a matrix is a) Number of columns b) Number of rows c) Number of non-zero rows in echelon form d) Determinant	1.00	CO1	L1
	2A2.	If $A$is invertible, then $\det(A)$is a) Zero b) One c) Non-zero d) Negative	1.00	CO1	L1
	2A3.	Row operations are mainly used to find a) Eigenvalues b) Inverse c) Echelon form d) Trace	1.00	CO2	L2
	2A4.	If $\lambda$is an eigenvalue of A, then eigenvalue of $3A$ is a) λ b) 3λ c) λ^2 d) $1/\lambda$	1.00	CO2	L2

	2A5.	Orthogonal matrices satisfy a) $A^T = A$ b) $A^T A = I$ c) $\det(A) = 0$ d) $A^2 = I$	1.00	CO2	L2
	2B.	Check whether the vectors (1,2,3), (2,4,6) are linearly independent.	3.00	CO1	L2
	2C.	Solve the following system using row reduction: x+y = 32 x+3y = 7	3.00	CO1	L3
	2D.	Apply the Gram–Schmidt process to obtain an orthonormal set from (1,1,0), (1,0,1).	7.00	CO2	L3
3.	3A1.	A matrix equal to its transpose is a) Orthogonal b) Skew-symmetric c) Symmetric d) Singular	1.00	CO1	L1
	3A2.	Determinant changes sign when a) A row is multiplied by a scalar b) Two rows are interchanged c) Rows are added d) Matrix is transposed	1.00	CO1	L1
	3A3.	Diagonalization of a matrix requires a) Zero determinant b) Repeated eigenvalues c) Linearly independent eigenvectors d) Orthogonal rows	1.00	CO2	L2
	3A4.	The dot product of orthogonal vectors is a) 1 b) –1 c) 0 d) Maximum	1.00	CO2	L1

	3A5.	Gram–Schmidt process is based on a) Determinants b) Eigenvalues c) Orthogonality d) Diagonalization	1.00	CO2	L2
	3B.	Find the dot product of vectors (1,2,3) and (4,5,6).	3.00	CO1	L3
	3C.	Let Abe a 3×3 matrix such that the second row of $A$is twice the first row. Without calculating the determinant directly, determine $\det(A)$ and justify your answer using the properties of determinants.	5.00	CO1	L3
	3D.	Determine whether the matrix $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ is diagonalizable.	7.00	CO1	L4

Model Question Paper

Question Paper Title: MINOR II SEM IV		
Total Duration (H:M): 1:30	Course: **Linear Algebra and Calculus** (AS2004)	Maximum Marks: 40
Note: Attempt ANY TWO questions from Q.1, Q.2 and Q.3		

Q. No.		**Questions**	**Marks**	**CO**	**BL**
1.	1A1.	The domain of the function $f(x) = \sqrt{x-2}$is: a) $x > 0$ b) $x \geq 2$	1.00	CO3	L1

		c) $x < 2$ d) All real numbers			
	1A2.	The limit $\lim_{x\to 0} \frac{\sin x}{x}$is: a) 0 b) 1 c) ∞ d) Does not exist	1.00	CO3	L1
	1A3.	The directional derivative of a function at a point gives: a) Rate of change in a direction b) Maximum value of function c) Gradient vector magnitude d) Zero	1.00	CO4	L2
	1A4.	A function $f(x)$is continuous at $x = a$if: a) $f(a)$exists b) $\lim_{x\to a} f(x)$exists c) $\lim_{x\to a} f(x) = f(a)$ d) None of the above	1.00	CO3	L1
	1A5.	If the Hessian determinant of $f(x, y)$at a critical point is negative, then the point is: a) Local maximum b) Local minimum c) Saddle point d) Not defined	1.00	CO4	L2
	1B.	Compute the directional derivative of $f(x, y) = x^2 + y^2$at point (1,2) in the direction of vector $\mathrm{u} = \frac{1}{\sqrt{2}}(1,1)$.	3.00	CO4	L3
	1C.	For $f(x, y) = x^2 + xy + y^2$, find the Hessian matrix at point (1,1).	5.00	CO4	L3
	1D.	Find the exact area bounded by the curve $y = x^2 + 2x$and the x-axis between $x = 1$and $x = 3$.	7.00	CO3	L3

2.	2A1.	The range of $f(x) = e^x$is: a) $(-\infty, \infty)$ b) $[0, \infty)$ c) $(0, \infty)$ d) $(-1,1)$	1.00	CO3	L1
	2A2.	The gradient vector of $f(x, y) = x^2 + y^2$at (1,1) is: a) (1,1) b) (2,2) c) (0,0) d) (1,0)	1.00	CO4	L1
	2A3.	Which of the following rules is used to differentiate a product of two functions? a) Chain rule b) Product rule c) Quotient rule d) Power rule	1.00	CO3	L2
	2A4.	The triple integral $\iiint_V 1\, dV$gives: a) Area b) Volume c) Surface d) Gradient	1.00	CO3	L2
	2A5.	For optimization problems, a point where $\nabla f = 0$ is called: a) Boundary point b) Stationary point c) Asymptote d) Singular point	1.00	CO4	L2
	2B.	Find the derivative of $f(x) = x^3 - 5x^2 + 6x - 1$and evaluate it at $x = 2$.	3.00	CO3	L2
	2C.	Find the gradient vector of $f(x, y) = xy + y^2$at the point (1,1).	5.00	CO4	L3

	2D.	Find the stationary points of $f(x,y) = x^4 + y^4 - 4xy + 1$. Use the gradient and Hessian to classify each point as maxima, minima, or saddle points.	7.00	CO4	L4
3.	3A1.	The derivative of a constant function is: a) 1 b) 0 c) Undefined d) Infinity	1.00	CO3	L1
	3A2.	The limit $\lim_{x\to\infty} \frac{1}{x}$is: a) 0 b) 1 c) ∞ d) Does not exist	1.00	CO3	L1
	3A3.	The Hessian matrix of $f(x,y)$is: a) Matrix of second derivatives b) Gradient vector c) Jacobian d) Zero	1.00	CO4	L2
	3A4.	The second derivative test for functions of two variables requires computing: a) Gradient b) Hessian determinant c) Partial derivatives only d) None of the above	1.00	CO4	L2
	3A5.	The double integral $\int_0^1 \int_0^x (x+y)\, dy\, dx$gives: a) Area b) Volume c) Gradient d) Derivative	1.00	CO3	L2
	3B.	Compute the double integral $\int_0^1 \int_0^y (x+y)\, dx\, dy$	3.00	CO3	L3

	3C.	Compute the directional derivative of $f(x,y) = x^2 + y^2$at point (1,2) in the direction of vector $\mathrm{u} = \frac{1}{\sqrt{2}}(1,1)$.	5.00	CO4	L3
	3D.	A solid region is bounded by $z = 0$, $z = x + y$, and $x = 0, y = 0, x + y = 1$. Compute the volume of the solid using a triple integral.	7.00	CO3	L4

Model Question Paper

Question Paper Title: END SEM **SEM IV**		
Total Duration (H:M): 3 Hours	Course: **Linear Algebra and Calculus** (AS2004)	Maximum Marks: 100
Instruction: Answer two questions from Unit I, two questions from Unit II, and one question from Unit III.		

Q. No.		**Questions**	**Marks**	**CO**	**BL**
		Unit I (Attempt any two of the following)			
1.	1A1.	If $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$, what is $\det(A)$? (a) 5 (b) 8 (c) 4 (d) 6	1.00	CO1	L1
	1A2.	Which of the following is an eigenvalue of $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$? (a) 1 (b) 3 (c) 2 (d) 0	1.00	CO2	L2
	1A3.	Which of the following matrices is upper triangular?	1.00	CO1	L1

		(a) $\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$ (b) $\begin{bmatrix}1 & 2\\0 & 3\end{bmatrix}$ (c) $\begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$ (d)$\begin{bmatrix}2 & 0\\1 & 2\end{bmatrix}$			
	1A4.	If $v_1 = (1,2,0)$and $v_2 = (2,-1,0)$, are they orthogonal? (a) Yes (b) No (c) Only if scaled (d) Cannot determine	1.00	CO2	L1
	1A5.	Which matrix is invertible? (a) $\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$ (b) $\begin{bmatrix}2 & 3\\1 & 2\end{bmatrix}$ (c) $\begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$ (d) $\begin{bmatrix}1 & 1\\2 & 2\end{bmatrix}$	1.00	CO1	L1
	1B.	Apply the Gram-Schmidt process to orthogonalize the vectors $v_1 = (1,0,1)$and $v_2 = (1,1,0)$.	3.00	CO2	L3
	1C.	Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix}3 & 1\\0 & 2\end{bmatrix}$.	5.00	CO2	L2
	1D.	Find the eigen vectors of $B = \begin{bmatrix}2 & -1\\-1 & 2\end{bmatrix}$ and explain why this matrix is diagonalizable.	7.00	CO2	L3
2.	2A1.	The inverse of the matrix $A = \begin{bmatrix}2 & 3\\1 & 2\end{bmatrix}$exists if: (a) det(A) = 0 (b) det(A) ≠ 0 (c) trace(A) = 0 (d) A is symmetric	1.00	CO1	L1
	2A2.	Which of the following sets of vectors are orthogonal? (a) $(1,0),(0,1)$ (b) $(1,1),(1,1)$ (c) $(2,3),(4,6)$ (d) $(1,0),(1,1)$	1.00	CO2	L2
	2A3.	If $A = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$, what is det(3A)? (a) 6 (b) 18 (c) 54 (d) 12	1.00	CO1	L2

	2A4.	The eigenvectors of a symmetric matrix are always: (a) Linearly dependent (b) Orthogonal (c) Zero (d) Cannot determine	1.00	CO2	L2
	2A5.	Which of the following matrices is singular? (a) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}$	1.00	CO1	L2
	2B.	Reduce the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}$ to row echelon form.	3.00	CO1	L2
	2C.	Solve the following system of linear equations using the inverse of a matrix method: $x + y + z = 6, 2x - y + z = 3, x + 2y - z = 2$	5.00	CO1	L2
	2D.	Determine if the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{bmatrix}$ is invertible and explain why its structure guarantees invertibility.	7.00	CO1	L3
3.	3A1.	What is the trace of the matrix $A = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}$? (a) 5 (b) 9 (c) 11 (d) 4	1.00	CO1	L1
	3A2.	Are the vectors $v_1 = (1,1,0)$and $v_2 = (0,1,1)$linearly independent? (a) Yes (b) No (c) Only if scaled (d) Cannot determine	1.00	CO2	L1
	3A3.	Which of the following matrices is diagonalizable? (a) $\begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$	1.00	CO2	L2

	3A4.	Find the determinant of $A = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$ (a) 5 (b) 6 (c) 3 (d) 4	1.00	CO1	L2
	3A5.	Which of the following matrices is invertible? (a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	1.00	CO1	L1
	3B.	Which of the following matrices is invertible? (a) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	3.00	CO1	L2
	3C.	Find the eigenvalues and eigenvectors of the matrix: $A = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ Determine if $A$is diagonalizable.	5.00	CO2	L3
	3D.	Use the **Gram-Schmidt process** to orthogonalize the vectors: $v_1 = (1,1,0), v_2 = (1,0,1), v_3 = (0,1,1)$ Write the resulting orthogonal vectors clearly.	7.00	CO2	L3
Unit II (Attempt any two of the following)					
4.	4A1.	If $f(x) = x^2 + 3x + 2$, what is $f'(x)$? (a) $2x + 3$(b) $x^2 + 3$(c) $2x + 2$(d) $3x + 2$	1.00	CO3	L1
	4A2.	The limit $\lim_{x \to 2} \frac{x^2-4}{x-2}$equals: (a) 0 (b) 2 (c) 4 (d) 5	1.00	CO3	L2

	4A3.	The gradient of $f(x,y) = x^2 + y^2$is: (a) $2x + 2y$ (b) $(2x, 2y)$ (c) (x^2, y^2) (d) $(x + y, x + y)$	1.00	CO4	L1
	4A4.	The Hessian matrix of $f(x,y) = x^3 + y^3$is: (a) $\begin{bmatrix} 6x & 0 \\ 0 & 6y \end{bmatrix}$ (b) $\begin{bmatrix} 6x & 0 \\ 0 & 6y \end{bmatrix}$ (c) $\begin{bmatrix} 6x & 0 \\ 0 & 6y \end{bmatrix}$ (d) $\begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$	1.00	CO2	L1
	4A5.	Which of the following functions is continuous at $x = 0$? (a) $f(x) = \frac{1}{x}$ (b) $f(x) = x^2$ (c) $\frac{1}{x^2}$ (d) $sinx$	1.00	CO1	L2
	4B.	Find the derivative of $f(x) = x^3 - 5x^2 + 4x - 1$.	3.00	CO3	L2
	4C.	Use derivatives to find the maximum and minimum of $f(x) = x^3 - 6x^2 + 9x + 1$ on the interval [0,4].	5.00	CO3	L3
	4D.	Determine the Hessian matrix, and classify the critical points of $f(x,y) = x^2 + xy + y^2 - 4x - 6y$ using gradient vectors.	7.00	CO4	L3
5.	5A1.	If $f(x) = \sqrt{x^2 + 1}$, what is $f'(x)$? (a) x(b) $\frac{x}{\sqrt{x^2+1}}$(c) $\frac{1}{2\sqrt{x^2+1}}$(d) $\sqrt{x^2 + 1}$	1.00	CO3	L1
	5A2.	The gradient of $f(x,y) = xy + y^2$ at the point $(1, 2)$is: (a) $(2,1)$ (b) $(2,5)$ (c) $(2,4)$ (d) $(1,1)$	1.00	CO4	L1
	5A3.	The limit $\lim_{x \to 0} \frac{\sin x}{x}$equals: (a) 0 (b) 1 (c) ∞(d) Does not exist	1.00	CO3	L2

	5A4.	The derivative of $f(x) = x^3$at $x = 2$is: (a) 4 (b) 6 (c) 8 (d) 12	1.00	CO2	L2
	5A5.	The Hessian matrix of $f(x, y) = x^2y + y^3$is: (a) $\begin{bmatrix} 2y & 2x \\ 2x & 6y \end{bmatrix}$ (b) $\begin{bmatrix} 2y & x \\ x & 6y \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 0 \\ 0 & 6 \end{bmatrix}$ (d) $\begin{bmatrix} y & x \\ x & y \end{bmatrix}$	1.00	CO4	L2
	5B.	Compute the gradient vector of $f(x, y) = x^2y + y^3$ at the point (1,2).	3.00	CO4	L2
	5C.	Evaluate the triple integral $\iiint_V z\, dV$over the cuboid $0 \le x \le 1, 0 \le y \le 2, 0 \le z \le 3$	5.00	CO3	L3
	5D.	For $f(x, y) = x^3 - 3xy^2$, find all critical points and determine their nature using second-order partial derivatives.	7.00	CO1	L3
6.	6A1.	The derivative of $f(x) = 5x^2$at $x = 3$is: (a) 15 (b) 30 (c) 45 (d) 60	1.00	CO3	L1
	6A2.	If $f(x, y) = x^2 + y^2$, the gradient at $(1, 2)$is: (a) (1,2) (b) (2,4) (c) (2,2) (d) (1,1)	1.00	CO4	L1
	6A3.	The limit $\lim_{x \to 1}(3x + 2)$is: (a) 3 (b) 4 (c) 5 (d) 6	1.00	CO3	L1
	6A4.	Which function is continuous at $x = 0$? (a) $f(x) = x^2$(b) $f(x) = \frac{1}{x}$(c) $f(x) = \frac{1}{x^2}$(d) $f(x) = \ln(x)$	1.00	CO3	L1
	6A5.	The Hessian of $f(x, y) = x^2 + xy + y^2$is:	1.00	CO4	L2

		(a) $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$(b) $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$(c) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}(d)\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$			
	6B.	Evaluate the double integral $\iint_R (x+y)\,dx\,dy$over the rectangle $R: 0 \le x \le 1, 0 \le y \le 2$.	3.00	CO3	L2
	6C.	Find the critical points of $f(x,y) = x^2 + xy + y^2$and determine their nature using the Hessian matrix.	5.00	CO4	L3
	6D.	Compute the volume under the surface $z = x^2 + y^2$over the square region $0 \le x \le 1, 0 \le y \le 1$using a double integral.	7.00	CO3	L3
Unit III (Attempt any one of the following)					
7.	7A1.	If the cost function is $J(\theta) = (\theta - 5)^2$, what is the value of θthat minimizes $J(\theta)$? (a) 0 (b) 2 (c) 5 (d) -5	1.00	CO5	L1
	7A2.	Which of the following is a **convex function**? (a) $f(x) = x^3$(b) $f(x) = x^2$(c) $f(x) = -x^2$(d) $f(x) = \sin x$	1.00	CO5	L1
	7A3.	Gradient descent is used to: (a) Maximize cost functions (b) Minimize cost functions (c) Compute eigenvalues (d) Solve linear systems	1.00	CO5	L1
	7A4.	For $J(\theta_1, \theta_2) = \theta_1^2 + \theta_2^2$, the gradient $\nabla J$is: (a) (1,1) (b) $(2\theta_1, 2\theta_2)$ (c) (θ_1^2, θ_2^2) (d) (0,0)	1.00	CO2	L1
	7A5.	In machine learning, minimizing the cost function helps to:	1.00	CO5	L2

		(a) Improve model accuracy (b) Increase error (c) Randomize predictions (d) None of the above			
	7B.	The cost function for a linear regression model is $J(\theta) = (\theta - 4)^2 + 1$. Find the value of θthat minimizes $J(\theta)$and calculate the minimum cost.	3.00	CO2	L3
	7C.	Consider the function $f(\theta_1, \theta_2) = (\theta_1 - 2)^2 + (\theta_2 + 3)^2$. Find the values of $\theta_1$and $\theta_2$that minimize $f$and determine the minimum value.	5.00	CO5	L3
	7D.	A machine learning cost function is given by $J(\theta_1, \theta_2) = \theta_1^2 + 2\theta_2^2 - 4\theta_1 - 8\theta_2 + 10$. Use gradient vectors to find the point (θ_1, θ_2)that minimizes $J$and compute the minimum cost.	7.00	CO5	L3
8.	8A1.	The cost function $J(\theta) = (\theta + 2)^2$ is minimized at: (a) 0 (b) -2 (c) 2 (d) 1	1.00	CO5	L1
	8A2.	A convex cost function has: (a) Multiple local minima (b) Only one global minimum (c) No minimum (d) Only maxima	1.00	CO5	L2
	8A3.	If the learning rate in gradient descent is too large, the algorithm may: (a) Converge faster (b) Overshoot and fail to converge (c) Always reach global minimum (d) Stop immediately	1.00	CO5	L2
	8A4.	Which of the following is an example of a cost function in ML? (a) $J(\theta) = \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$(b) $f(x) = x^3$(c) $g(x) = \sin\ x$(d) $h(x) = x + 1$	1.00	CO5	L1

	8A5.	In ML optimization, the purpose of regularization is to: (a) Increase training error (b) Reduce overfitting (c) Remove derivatives (d) Randomly change parameters	1.00	CO5	L2
	8B.	The cost function $J(\theta) = 2(\theta - 3)^2 + (\theta - 5)^2$ Find the value of θ that minimizes J and the minimum cost.	3.00	CO1	L2
	8C.	A machine learning model has the cost function $J(\theta) = (\theta - 3)^4$. Determine if the function is convex and find the value of θ that minimizes J.	5.00	CO5	L3
	8D.	Suppose a linear regression model uses the cost function $J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^{m} (h_\theta(x^{(i)}) - y^{(i)})^2$. Explain conceptually how gradient descent updates $\theta_0$and θ_1, why the learning rate matters, and what happens if the function is non-convex.	7.00	CO5	L3

Sl. No.	Author/s	Title of the book/Digital resource	Publisher and Edition/URL
1.	Gilbert Strang	Linear Algebra and Its Applications	Cengage Learning, 5th Edition
2.	Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence	Linear Algebra	Pearson, 5th Edition
3.	James Stewart	Calculus: Early Transcendentals	Cengage Learning, 8th Edition
4.	Boyd, S., & Vandenberghe, L.	Convex Optimization	Cambridge University Press, 1st Edition

Course Plan

Semester: 04 Year: 2024-25

Course Title: **Object Oriented Software Engineering**	Course Code: **SC2401**
L-T-P : 3-1-0	Credits: 04
Total Contact hrs: 60	Duration of ESA: 3 hours
ISA Weightage: 50	ESA Weightage:50
Lesson Plan Authors: Dr. Preethi Baligar	Date:
Checked By: Dr. Savita Hanji and Dr. Stuti Ahuja	Date:

Prerequisites:

No Pre-requisite

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Identify the need to engineer a software system.
2. Describe Software Engineering Lifecycle Models
3. Perform software requirements analysis
4. Apply knowledge of the System Analysis and Design concepts using UML.
5. Apply software testing and maintenance approaches
6. Use basic project management practices and tools for software development including estimation

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title : **Object Oriented Software Engineering**	Semester: IV
Course Code: **SC2401**	Year: 2024

Course Outcomes (COs) / Program Outcomes (POs)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
Reference from Program Articulation Matrix														
1. Identify the need to engineer a software system.	H													
2. Describe Software Engineering Lifecycle Models	H													
3. Perform software requirements analysis		H												
4. Apply knowledge of the System Analysis and Design concepts using UML.	H				L									
5. Apply software testing and maintenance approaches	M													
6. Use basic project management practices and tools for software development including estimation					L									

Degree of compliance **L**: Low **M**: Medium **H**: High

Course Content

Course Code: Object Oriented Software Engineering	Course Title: **SC2401**	
L-T-P : 3-1-0	Credits: 03	Contact Hrs: 60
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 60		Exam Duration: 03 hours

Content	Hrs
Unit - 1	
1. Introduction to Software Engineering: Need for and Introduction to Software Engineering and A Generic view of process, Software engineering ethics	5 hrs
2 Process Models: Prescriptive Models, The waterfall model, Incremental process models, Evolutionary process models, Specialized process models, The Unified process; Agile view of process: principles, frameworks and practices; Scrum	8 hrs
3 Requirements Engineering: Requirements Engineering tasks, Initiating Requirements Engineering Process: the groundwork; Requirements gathering, Elicitation Work Products, Developing Use-Cases , Analysis Model, Negotiating Requirements and Validating requirements;	8 hrs
Unit - 2	
4. Pattern based software design, Architectural design: Software Design: Engineering Design within the context of SE, Design process and design quality, Design concepts, the design Model, Architecture, Data design, Architectural styles and patterns, Architectural design	6 hrs
5 Overview of object-oriented concepts Unified Modeling Language (UML). Class Model, State Model and Interaction Models: Use case, sequence and activity diagrams.	10 hrs
6 Object Oriented System Design: Reuse Plan, breaking a system into sub-systems and organizing. Allocation of sub-systems to hardware and software. High Level Class Design: Design Optimization, Adjustment of Inheritance and Organizing a class design.	10 hrs
Unit - 3	

7 Testing Strategies: A strategic approach to software testing, Test strategies for conventional software, validation testing, system testing. Testing tactics: White box testing, basis path testing, control structure testing, black box testing, testing for specialized environments, architectures and applications.	7 hrs
8 . Project Management and Metrics: Management spectrum, The people, product, process , metrics in the process and project domains, software measurements, metrics for software quality. Project Estimation: Observations on estimation, the project planning process , software scope and feasibility resources, software project estimation, Decomposition techniques, empirical estimation models, DevOps: introduction, Tools and practices; Tools- CaseStudy	6 hrs

Learning Resources:

1. Roger S Pressman, Software Engineering - A Practitioner Approach, Seventh Edition, McGrawHill International Edition, 2009
2. Ian Sommerville, Software Engineering, Seventh Edition, Pearson education, 2004.
3. Blaha M, Rumbaugh, Object Oriented Modeling and Design with UML, Second, Pearson, 2008
4. Ali Bahrami, Object Oriented System Development using U M Languages, McGrawHill, 2008

References

1. Timothy C. Lethbridge and Robert Laganière, Object-Oriented Software Engineering: Practical Software Development using UML and Java , McGraw-Hill Education
2. Grady Booch et al., Object-Oriented Analysis and Design with Applications, Addison-Wesley Professional
3. Erich Gamma et al, Design Patterns: Elements of Reusable Object-Oriented Software, Addison-Wesley Professional
4. Craig Larman Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development" by Pearson Education

CIA Evaluation Scheme*

Assessment	Weightage in Marks
Minor Exam-1 on Unit 1 portion	20*
Minor Exam-2 on Unit 2 portion	20*
Assignment(s)/MOOC Certification on DevOps/Application to Mini project	10
Total	50

*For theory courses. Change of wight to 15-15-20, if desired, requires prior written approval from Dean of School, before the commencement of semester.

#For Lab / Project courses the weightage of (ISA : ESA) is (80 : 20). The 80% assessment has to be done in multiple (not less than 05) occasions / phases / sessions

Course Assessment Plan

Course Title: Object Oriented Software Engineering Code: **SC2401**						
Course outcomes (COs)	Weightage in assessment	Assessment Methods				
		Minor 1 (Written)	Minor 2 (Written)	Activity	ESA (Written)	
1. Identify the need to engineer a software system.		10			10	
2. Describe Software Engineering Lifecycle Models		15			15	
3. Perform software requirements analysis		15			15	
4. Apply knowledge of the System Analysis and Design concepts using UML.			24		24	
5. Apply software testing and maintenance approaches			16		16	
6. Use basic project management practices and tools for software development including estimation				10	20	
Weightage		40 (20% weight)	40 (20% weight)	10 (10% weight)	100 (50% weight)	

Date: **Program Head**

Chapterwise Plan

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: 1. **Introduction to Software Engineering**	Planned Hours**: 5 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Recognize the importance of structured processes in software development	CO1	L2	1
2. Define Software Engineering and its need in designing software systems.	CO1	L2	1
3. Explain software engineering practice	CO1	L2	1
4. Explain the generic view of software process	CO1	L2	1
5. Explain the importance of ethics and professional responsibilities in software development	CO1	L2	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Overview view of the course
2. Introduction to software engineering
3. Need for software Engineering
4. A generic view of software process
5. Software engineering ethics

Review Questions

Sl.No. - Questions	TLOs	BL
1. What challenges arise in software development without a structured engineering process? How does a structured process improve the quality and reliability of software?	TLO1	L2

2. What is the definition of software engineering, and how does it differ from traditional programming?	TLO2	L2
3. What is the need of a software engineering? Why is it required?	TLO3	L2
4. With the help of a neat block diagram, explain the generic view of a software development process.	TLO4	L2
6. Why are ethics and professional responsibilities crucial in software development?	TLO5	L2

Chapter wise Plan

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: 2.Process Models	Planned Hours: 08 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the different prescriptive and specialized process models used for software development	CO2	L2	1
2. Discuss the need for agile methodology in comparison with prescriptive process models	CO2	L2	1
3. Discuss how XP can be implemented in an Agile process to improve collaboration, code quality, and adaptability in projects.	CO2	L2	1
4. Plan a given software using SCRUM methodology	CO2	L3	1
5. Describe how a Kanban board is used to manage tasks and optimize workflow in a project.	CO2	L2	1
6. Explain the objectives and benefits of DevOps, focusing on its role in bridging development and operations teams.	CO2	L2	1
7. Choose an appropriate process model based on the characteristics of the mini project/scenario	CO2	L4	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Prescriptive Process models: The waterfall model, Incremental Process Model
2. Prescriptive Process Models (continued) : Evolutionary and concurrent process models
3. Specialized Process Models
4. The unified process model
5. Agile Process: SCRUM
6. Agile Process Models: Extreme Programming

7.Other Agile frameworks: Kanban, DevOps

Review Questions

Sl.No. - Questions	TLOs	BL
1. Compare and contrast two process models (e.g., Waterfall vs. Incremental) for a project where frequent changes in requirements are expected. Which model would you apply, and why?	TLO1	L3
2. A project with a high degree of uncertainty in requirements and a need for early prototyping is starting. How would you apply an evolutionary process model to ensure success in the early stages of development?	TLO2	L3
3. You have been assigned to a project that involves iterative development. How would you apply the Unified Process phases to manage iterations and align them with customer feedback?	TLO3	L3
4. You are managing a project that requires continuous customer feedback and frequent releases. How would you apply the principles of extreme programming to ensure alignment with customer needs and minimize rework?	TLO5	L4
5. Take the case the Instagram/chatgpt/e-commerce applications/word processor. Compare the merits and demerits of its implementation using at least three process models of your choice	TLO6	L3
6. Compare agile software development model with any other two models of your choice.	TLO5	L2
7. Which of the development process models discussed would you follow for the following projects and justify your answers. i. An on-line store. ii. A Toll Collection System for Highway Authorities to collect Toll at multiple entry points between PUNE and HUBLI. iii. A new operating system for Mobile Devices to reach market in defined amount of time which is short	TLO6	L3

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: **3. Requirements engineering**	Planned Hours: 8 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain requirements engineering process	CO3	L2	2
2. Elicit requirements of a system under consideration	CO3	L4	2
3. Develop use-cases of a given system	CO3	L3	2
4. Build requirements model for a system	CO3	L3	2
5. Explain the importance of negotiation in requirements engineering	CO3	L2	2
6. Apply requirements engineering to the mini project need statement	CO3	L3	2

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to requirements engineering (RE)
2. Pre-requisites for requirements engineering
3. Requirements specification
4. Functional and non-functional requirements
5. Developing Use Cases
6. Building the analysis model
7. Negotiating and validating requirements
8. Tutorial : requirements analysis

Review Questions

Sl.No. – Questions	TLOs	BL
1. Define the concept of requirements engineering and explain its role in software development. Given a project with diverse stakeholders, explain how you would apply the requirements engineering process to manage their different expectation	TLO1	L3
2.Given a project with unclear stakeholder needs, explain how you would apply the elicitation techniques to gather comprehensive and precise requirements	TLO2	L3
3. Develop a complete use-case for making withdrawal at an ATM	TLO3	L3
4.Elicit functional and non-functional requirements of a coffee vending machine.	TLO2	L2
5. How can software engineers identify stakeholders during inception?	TLO2	L2
6. How do primary and secondary actors differ from one another in use-case descriptions?	TLO3	L3
7 Write atleast six functional requirements for the e-commerce Amazon's ecosystem, focusing on a seamless integration with social media, entertainment, and personalization for Gen Z users. The goal is to create an immersive shopping experience that leverages short-form video content, influencer recommendations, and community interaction, all within the Amazon environment.	TLO4	L4
8. In a project with conflicting stakeholder requirements, explain how you would apply negotiation techniques to reach a consensus and ensure all requirements are agreed upon.	TLO5	L3

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: 4. **Software Design**	Planned Hours: 6 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the nature of the design process	CO4	L2	1
2. Apply design concepts in software engineering	CO4	L3	1
3. Explain the major categories of software architectural styles.	CO4	L2	1
4. Choose an appropriate architectural style for a given software project	CO4	L4	1
5. Describe the design of a system choosing appropriate representation form	CO4	L3	1
6. Apply suitable design patterns to address common software design problems in a given scenario	CO4	L3	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. The software design process, Design concepts
2. Design model
3. Software Architecture
4. Architectural styles, Architectural Design
5. Pattern-based software design

Review Questions

Sl.No. - Questions	TLOs	BL
Describe the concept of modularity in software design. Why is it important for maintainability and scalability?	TLO1	L2
1.What are the primary categories of software architectural styles, and what are their key characteristics?	TLO3	L2

2.Analyze a system that handles real-time data processing (e.g., youtube streaming services). Which architectural style would best suit this application, and how does it meet the system's requirements?	TLO4	L4
3.For the following systems, identify what you consider to be their major features and identify the architectural styles that are likely to be most appropriate: a) A bank auto-teller machine b) A spell-checker used to analyse static files of text c) A program that reformats „raw text" into the page description language employed by a particular printer.	TLO4	L4
4. Design a simple banking system. Choose the most appropriate representation forms to capture key components such as user interactions, transactions, and security.	TLO5	L3
5. In the context of an online store, apply the Factory design pattern to create a product catalog. Explain how this pattern facilitates product creation and reduces coupling between system components	TLO6	L3
6. Given a large software project, how would you apply the design concepts of modularity and abstraction to divide the system into manageable components? Describe how these concepts would help in improving system scalability and reducing complexity	TLO2	L3

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: 5. **Overview of object-oriented concepts**	Planned Hours: 10 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain object-oriented concepts and the need to develop different models for a system	CO4	L2	1
2. Describe the components and structure of the Unified Modeling Language (UML) and its role in representing system architecture	CO4	L2	1
3. Apply UML to create class diagrams, representing object relationships and attributes of a system	CO4	L3	1
4. Use UML to create state models, representing transitions between different states of the system.	CO4	L3	1
5. Develop sequence diagrams to illustrate object interactions and time sequences for specific use cases within a system.	CO4	L3	1
6. Develop activity diagrams to model the workflow of processes, representing control flow and decision points in a system	CO4	L3	1
7. Develop class and interaction diagrams as appliable to the project need statement	CO4	L4	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Overview of object-oriented concepts.
2. Modelling as a design technique
3. Unified Modeling Language (UML)
4. Class Model
5. State Model
6. Sequence diagrams

7. Activity diagrams
8. Application of modelling to the mini project need statement

Review Questions

Sl.No. - Questions	TLOs	BL
1. Discuss the need for Object oriented system design for software development.	TLO1	L2
2. What are the key components of the Unified Modeling Language (UML), and how does UML help in representing the system architecture?	TLO3	L2
3. Design a Class Model for a Music Streaming App that includes the following functionalities: a. Subscription management (handling user subscriptions to different tiers of service), b. Playlist creation and management (users can create, add, and remove songs from playlists), c. Song streaming and recommendations (users can stream songs and receive song recommendations based on their listening history and preferences). Use UML to create the class model representing objects, their relationships, and attributes in this music streaming app.	TLO3	L3
4. Using UML, design a state diagram for a user account in an online banking system, illustrating the different states (e.g., active, suspended, closed) and transitions between them.	TLO4	L3
5. Create a UML sequence diagram that represents the flow of messages during the login process for a web-based application.	TLO5	L3
6. Using UML, design an activity diagram to model the process of liking and commenting on a post in Instagram. The diagram should include the following: a. User Scrolls Feed: The user browses through their Instagram feed. b. User Likes Post: The user taps the heart icon to like a post. c. Decision Point: The user decides whether to leave a comment. d. User Comments (Optional): If the user decides to comment, they type and submit a comment.	TLO6	L3

e. End Process: The user either moves on to another post or closes the app.		

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: 6. **Object Oriented System Design**	Planned Hours: 10 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Illustrate the key principles of system design and its components with suitable examples.	CO4	L2	1
2. Modularize a given system into subsystems: Layers and Partitions	CO4	L3	1
3. Develop a reuse plan by identifying reusable components and defining strategies for their integration into a system.	CO4	L3	1
4. Identify and classify sub-systems for allocation to hardware and software	CO4	L2	1
5. Optimize the given class design	CO4	L5	1
6. Make Adjustment of Inheritance to fine tune the class design	CO4	L3	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Reuse Plan
2. Breaking a system into sub-systems
3. Breaking a system into sub-systems and organizing
4. Allocation of sub-systems to hardware and software
5. High Level Class Design: Design Optimization
6. Adjustment of Inheritance

7. Organizing a class design
8. Case studies for exploration and application

Review Questions *

Sl.No. - Questions	TLOs	BL
1. Discuss System design steps.	TLO1	L2
2. Discuss the approaches used for reuse in a system design.	TLO2	L2
3. How to break a system into subsystems	TLO2	L3
4. Organize the given subsystem into layers and partitions	TLO2	L3
5. Explain design optimization steps	TLO4	L2
7. Organize the classes to produce fine-tuned class model	TLO5	L3

*Review questions will be based on given case studies/mini project need statement

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: **7. Testing Strategies**	Planned Hours: 7 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the generic characteristics of software testing strategies	CO5	L2	1
2. Differentiate between verification and validation	CO5	L2	1
3. Apply white-box testing techniques to analyze software code	CO5	L3	1
4. Apply black-box testing techniques to evaluate system behavior	CO5	L3	1
5. Develop the test plan using appropriate strategies for testing a software	CO5	L3	1

Lesson Schedule

Class No. - Portion covered per hour / per Class
1. A strategic approach to software testing
2. Test strategies for conventional software
3. Validation testing, system testing
4. White box testing, basis path testing, control structure testing
5. Black box testing, testing for specialized environments, architectures and applications

Review Questions

Sl.No. - Questions	TLOs	BL
1. Using your own words, describe the difference between verification and validation. Do both make use of test case design methods and testing strategies?	TLO2	L2
2. Why is a highly coupled module difficult to unit test?	TLO2	L2
3. Give at least three examples in which black-box testing might give the impression that everything is fine , while white-box tests might uncover an error and vice versa.	TLO2	L3
4. Why do we have to retest subclasses that are instantiated from an existing class, if the existing class has already been thoroughly tested? Can we use the test cases designed for the existing class	TLO3	L2

Course Code and Title: **SC2401/ Object Oriented Software Engineering**	
Chapter Number and Title: **8. Project Management and Metrics**	Planned Hours: 6 **hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain how people, process and problems are managed during a software project	CO6	L2	7
2. Explain how software metrics be used to manage a software project and software process.	CO6	L2	7

3. Discuss the design metrics for conventional software and object-oriented software	CO6	L2	7
4. Explain the different estimation techniques in software project.	CO6	L2	7
5. Use Project management tools to manage software projects effectively	CO6	L3	7

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Management spectrum, The people, product, process , metrics in the process and project domains
2. software measurements, metrics for software quality
3. Observations on estimation, the project planning process
4. Software scope and feasibility, resources
5. Software project estimation
6. Decomposition techniques, empirical estimation models
7. DevOps: introduction, Tools and practices; Tools- CaseStudy

Review Questions

Sl.No. - Questions	TLOs	BL	PO
1. Explain how processes like planning and execution contribute to managing software projects.	CO6	L2	7
2. How can metrics help in tracking the progress and quality of a software project?	CO6	L2	7
3. Explain how metrics like reliability, maintainability, and usability are applied in evaluating software quality.	CO6	L2	7
4. How does the complexity of a project affect the composition and organization of the team?	CO6	L2	7
5. Show how DevOps tools enhance project management efficiency in software projects.	CO6	L3	7

Question Paper Title: MINOR I EXAMINATION		
Total Duration (H:M): 1:30	Course: SC2401/ Object Oriented Software Engineering	Maximum Marks: 40
Note: There are 3 questions, each of 20 marks, in the paper. Attempt any two of the 3 questions.		

Q	Q	Question	BL	CO	Marks
1	A	A process framework establishes the foundation for a complete software engineering process. Justify this statement.	L2	CO 1	5
	B	Which development process model would you choose for the following projects. Justify. i. A mobile app for online shopping that includes personalized recommendations and quick payment options. ii. An app-based toll collection system for highways between Pune and Solapur, allowing seamless payment through digital wallets. You are expected to identify at least two characteristics for each of the software which ideally suits the choice of a particular process model.	L3	CO 2	7
	C	Write at least six functional requirements for one of the software mentioned in Q 1(b). Choose one of these functional requirements and write two verifiable non-functional requirements related to this chosen functional requirement.	L3	CO 3	8
2	A	Explain how professional ethics influence decision-making in software development. Provide an example of an ethical dilemma and how it could be addressed.	L2	CO 1	5
	B	Apply SCRUM process knowledge and propose the project flow for any of the software mentioned in Q1 (b)	L3	CO 2	7
	C	You are designing the user registration process for an e-commerce platform like Amazon. The platform has various modules like User Registration, Product Browsing, and Order Placement. For any two modules of your choice-	L3	CO 3	8

		i) List the actors involved in it ii) Draw a Use Case Diagram for the chosen modules, iii) Provide a detailed Use Case Description for one of the modules			
3	A	How do the umbrella activities of the software engineering process framework help a software team manage and control progress, quality, change and risk?	L2	CO 1	5
	B	You are asked to build automation software for a university which includes the following modules: Students, Admission, Examination, and Attendance. Discuss the pros and cons of building this software using: i) Waterfall model ii) Unified Process model	L3	CO 2	7
	C	Describe use case diagram for “WhatsApp video call”, consisting of at least two actors and 4 scenarios. Describe one of the use cases from your list.	L3	CO 3	8

Question Paper Title: MINOR II EXAMINATION		
Total Duration (H:M): 1:30	Course: SC2401/ Object Oriented Software Engineering	Maximum Marks: 40
Note: There are 3 questions, each of 20 marks, in the paper. Attempt any two of the 3 questions.		

Q	Q	Question	BL	CO	Marks
1	A	Describe the concept of modularity in software design. Why is it important for maintainability and scalability? Provide two examples to support your explanation.	L2	CO4	5
	B	Select suitable architectural pattern and propose an architecture for any one of the following application. Justify your selection. i) Road Detection system from Satellite images. ii) SHAREIt – a file transfer software tool	L3	CO4	7
	C	Draw Activity diagram for any one of the application from Q1b	L3	CO4	8
2	A	Explain the need for design patterns in software engineering. How do design patterns help in solving common design problems? Provide an example of a design pattern used in software development, explaining how it addresses a specific problem	L2	CO4	5
	B	Analyze a system that handles real-time data processing (e.g., youtube streaming services). Which architectural style would best suit this application, and how does it meet the system's requirements?	L3	CO4	7
	C	For youtube file sharing application from Q2b and draw its sequence diagram.	L3	CO4	8
3	A	What are the primary categories of software architectural styles, and what are their key characteristics?	L2	CO4	5
	B	Consider a ride-sharing application that processes real-time data for ride requests, driver availability, and route optimization. What architectural style would you recommend for such a system? Justify your choice by explaining how it addresses the	L3	CO4	7

		application's scalability, real-time responsiveness, and fault tolerance.			
	C	Consider Road accident detection system; identify at least three objects along with attributes and operations. Draw the corresponding standard UML diagram to represent the same.	L3	CO4	8

Question Paper Title: END SEMESTER EXAMINATION					
Total Duration (H:M): 3.00		Course: SC2401/ Object Oriented Software Engineering		Maximum Marks: 100	
Note: Solve 2 out of questions 1,2 and 3; Solve 2 out of 4,5 and 6; Solve 1 out of 7 and 8					
Q	**Q**	**Question**	**BL**	**CO**	**Marks**
UNIT 1					
1	A	Explain the significance of a software process framework in ensuring the success of a software development project. How does it facilitate consistency, quality, and repeatability in the development process?	L2	CO1	5
	B	Select an appropriate development process model for the following projects and justify your choice. i. A social media application focused on image and video sharing. ii. A cloud-based file storage and sharing service, designed to scale rapidly and handle large amounts of data.	L3	CO2	7

		Identify at least two key characteristics for each software and justify why a particular process model would best suit the project's needs.			
	C	Write six functional requirements for one of the software mentioned in Q1(b). For one of these functional requirements, propose two verifiable non-functional requirements.	L3	CO3	8
2	A	How do ethics impact decision-making in the field of software development? Illustrate with an example of an ethical issue that might arise during a project, and explain how it can be ethically resolved.	L2	CO1	5
	B	Explain the SCRUM methodology and outline how it could be applied to manage the development process of one of the software projects mentioned in Q1(b).	L3	CO2	7
	C	Design the user registration process for a new mobile application like WhatsApp. The app has features like user registration, profile creation, and messaging. For any two features of your choice: i) List the actors involved in the process. ii) Create a Use Case Diagram for the chosen features. iii) Write a detailed Use Case Description for one of the features.	L3	CO3	8
3	A	How do umbrella activities within the software engineering process framework help teams in managing progress, controlling quality, and minimizing risks? Discuss the relationship between these activities and successful software delivery.	L2	CO1	5
	B	Consider the task of creating a software automation system for a university, which includes the following modules: Student Information, Course Management, Grading, and Notifications. Discuss the advantages and disadvantages of using:	L3	CO2	7
	C	Design a Use Case Diagram for a “Video Call” feature in a messaging app like WhatsApp. The diagram should include at least two actors and cover four different scenarios (e.g., initiating a video call, receiving a call, ending the call, and managing call	L3	CO3	8

		settings). Provide a detailed Use Case Description for one of these scenarios.			
UNIT II					
Q	**Q**	**Question**	**BL**	**CO**	**Marks**
4	A	Describe any 3 design concepts of software design. Why are they important for maintainability and scalability?	L2	CO4	5
	B	Choose an appropriate architectural pattern and propose a suitable architecture for any one of the following applications: i) A real-time traffic monitoring system. ii) An online video streaming platform like Netflix. Justify your selection of the architectural pattern based on scalability, reliability, and performance requirements.	L3	CO4	7
	C	Draw an Activity diagram for the "real-time traffic monitoring system" described in Q4B. Include at least five key activities and their transitions.	L3	CO4	8
5	A	Describe the importance of design patterns in addressing recurring software design problems. Provide an example of the Observer pattern and explain how it is useful in a social media application.	L2	CO4	5
	B	Analyze the requirements of a system that processes real-time data for online gaming services. Which architectural style would be best suited to this application, and why? Discuss how this style supports scalability, fault tolerance, and real-time responsiveness.	L3	CO4	7
	C	For a file-sharing application similar to SHAREit, draw a sequence diagram that shows the interactions between a user and the system for transferring a file. Explain the interactions and their sequence.	L3	CO4	8
6	A	What is Architectural design? Give the Advantages of explicit architecture	L2	CO4	5
	B	Consider a smart home automation system that controls devices like lights, thermostats, and security cameras. Which architectural style would you recommend for this system? Justify your recommendation by explaining how it addresses scalability, real-time data processing, and fault tolerance.	L3	CO4	7

	C	Identify three key objects in a Fastag detection system and list their attributes and operations. Create a UML class diagram to represent these objects and their relationships.	L3	CO4	8
Unit III					
7	A	Discuss the management spectrum of software projects, including the roles of people, product, and process in ensuring project success.	L2	CO5	5
	B	Discuss the observations and challenges typically faced during the software project estimation process. What are the main aspects that need to be considered during project planning?	L3	CO5	7
	C	An online Journal portal system need to be developed to manage subscriptions for readership, online submission of papers from authors, online review of papers from experts. Write possible test cases using test driven approach.	L3	CO5	8
8	A	What is the role of metrics in the project and process domains of project management? Discuss how software quality metrics can influence decisions in a software project.	L2	CO6	5
	B	Show how DevOps tools enhance project management efficiency in software projects.	L3	CO6	7
	c	The client requests new features every two weeks with a short turnaround time. The project needs to adapt quickly to these evolving requirements while ensuring high-quality releases. How would you leverage DevOps practices like automated testing, monitoring, and feedback loops to ensure that the new features are delivered rapidly without compromising software quality? How do these practices help improve team collaboration and streamline project management throughout the development and release cycle?	L3	CO6	8

Course Plan

Semester: 04 Year: 2025-26

Course Title: Advanced Database Concepts	Course Code: SC2113
L-T-P: 3-0-0	Credits: 04
Total Contact hrs: 45	Duration of ESA: 3 hours
ISA Weightage: 50	ESA Weightage:50
Lesson Plan Authors: Dr. Savita S. Hanji	Date:
Checked By:	Date:

Course Outcomes (COs)

At the end of the course, the student should be able to:

1. Demonstrate the concepts in relational database management systems and need for advanced data models
2. Apply appropriate aggregate-based data models for given real-world applications.
3. Analyse data models and databases for scale.
4. Analyse solutions for schema migration, and multi-model types.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Advanced Database Concepts	Semester: 4
Course Code: SC2113	Year: 2024-2025

Course Outcomes (COs) / Program Outcomes (POs)	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
Reference from Program Articulation Matrix														
Demonstrate the concepts in relational database management systems and need for advanced data models	M													
Apply appropriate aggregate-based data models for given real-world applications.	M													
Analyse data models and databases for scale.		L												
Analyse solutions for schema migration, and multi-model types.			L								L			

Degree of compliance **L**: Low **M**: Medium **H**: High

Course Content

Course Code: SC2113	Course Title: Advanced Database Concepts	
L-T-P : 3-0-0	Credits: 04	Contact Hrs: 60
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 60		Exam Duration: 03 hours

	Content	Hrs
	UNIT- I	
1.1	**Relational databases:** Introduction to DBMS, ER Model, ER Diagram, Overview of RDBMS systems, Relational Algebra, Structured Query Language, Shortcomings of relational databases	12
1.2	**Aggregate-based data models:** Aggregate data models: Key-Value Databases, Document Databases, Column-Family stores , Non-aggregate Data models.	13
	UNIT – II	
2.1	**Implementations of aggregates:** Riak, MongoDB, Cassandra, Modelling data for access	12
2.2	**Capturing relations** : Data with relations, Graph databases, Neo4j **At scale**: Distribution Models, Consistency, Version stamps, Map-Reduce, Deployment, Cluster configuration	13
	UNIT- III	
3.1	**As systems**: Schema migration, Polygot Persistence, Multi-model types, Beyond NoSQL, Choosing your database	10

Learning Resources:

S.No.	Author/s	Title of the book/Digital resource	Publisher and Edition/URL
01	Ramez Elmasri and Shamkant B. Navathe	**Relational databases**: "Fundamentals of Database Systems"	Pearson, Edition: 7th Edition (latest available), ISBN-10:

			0133970779, ISBN-13: 978-0133970777
02	Pramod Sadalge, Martin Fowler	NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence	Addison-Wesley Professional

Course Assessment Plan

	Assessment	**Weightage in Marks**
ISA Scheme (50 Marks)	Minor 1	15
	Minor 2	15
	Case Study	10
	Course Project	10
	ISA Total	**50**
ESA Scheme (50 Marks)	**Exam**	50
Total		**100**

Course outcomes (COs)	**Weightage in assessment**	**Minor 1 (Written)**	**Minor 2 (Written)**	**Activity**	**ESA (Written)**
Explain the need for advanced database management systems.	10	5			10
Apply appropriate aggregate-based data models for given real-world applications.	40	10			25
Analyse data models and	40		15		35

databases for scale.					
Analyse solutions for schema migration, and multi-model types.	10				30
Weightage		40	40	20	100

Date: **Program Head**

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 1. **Relational databases**	Planned Hours: 12

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BLL	PO No
6. Explain the basic concepts of a Database Management System (DBMS), including its purpose and applications.	CO1	L2	1
7. Demonstrate an ER diagram for a given database scenario, specifying entities, attributes, and relationships.	CO1	L3	1
8. Apply relational algebra operations to manipulate and query relational data.	CO1	L3	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Birdseye view of the course
2. Introduction to DBMS
3. ER Model, ER Diagram
4. Overview of RDBMS systems
5. Relational Algebra
6. Case study on Relational Algebra

Review Questions

Sl.No. - Questions	TLOs	BL
1. Demonstrate how a Database Management System (DBMS) solves the problems associated with traditional file processing systems by providing a real-world example.	1	1
2. Design an ER diagram for a university system to manage student registrations, where students can enroll in multiple courses taught by different professors. Include attributes and relationships.	2	1

3. Write and demonstrate a relational algebra expression to retrieve the names of all employees working in a specific department from a given database schema.	3	1

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 2. Structured Query Language	Planned Hours: **6 hrs**

Learning Outcomes :-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Demonstrate SQL statements to define and modify the database structure using DDL commands.	CO1	3	1
2. Demonstrate SQL queries to manipulate and retrieve data using basic DML and SELECT commands.	CO1	3	1
3. Demonstrate SQL queries using JOIN operations to combine data from multiple tables.	CO1	3	1
4. Explain the purpose of constraints and how they ensure data integrity.	CO1	2	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Basics of SQL
2. Data Definition Language,
3. Data Manipulation and Retrieval
4. Complex SQL Queries and Constraints
5. Understanding Relational Database Limitations
6. Alternative Database Models

Review Questions

Sl.No. - Questions	TLOs	BL
1. A library has two tables: • Books (BookID, Title, Author, Genre, Price) • Borrowers (BorrowerID, Name, BookID, BorrowDate) **Question**: Write an SQL query to create tables for above schema and retrieve the names of borrowers who have borrowed books in the "Science Fiction" genre. Use a join between the Books and Borrowers tables.	1	3
2. A company manages employees and their projects in two tables: • Employees (EmpID, Name, Department, Salary) • Projects (ProjectID, EmpID, ProjectName, Deadline) **Question**: Write SQL queries to: • Insert a new employee and assign them to a project. • Update the deadline of a project for a specific employee. • Retrieve the names of employees working on projects with deadlines in the next 30 days.	2	3
3. An e-commerce website maintains two tables: • Customers (CustomerID, Name, City) • Orders (OrderID, CustomerID, OrderDate, TotalAmount) **Question**: Write an SQL query to retrieve the names of customers who have placed orders totaling more than 5000 in the last month. Ensure the query uses a join and appropriate date functions.	3	3
4. A streaming service manages its data using two tables: • Users (UserID, Name, SubscriptionPlan) • WatchHistory (WatchID, UserID, MovieName, WatchDate) **Question**: Explain how relational databases handle this data and demonstrate a query to retrieve the names of users who have watched more than 5 movies in the past week. Additionally, discuss why a NoSQL database might be more efficient for storing and querying this type of data.	4	3

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 3. Aggregate Data Model	Planned Hours: **13 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the concept of aggregate-based data models and their importance in distributed systems.	2	2	1
2. Design a simple key-value database for a given application.	2	3	1
3. Explain the structure and querying techniques of aggregate data models.	2	2	1

Lesson Schedule Class No. - Portion covered per hour / per Class
Introduction to Aggregate-based Data Models
Key-Value Databases
Document Databases
Column-Family Stores
Introduction to Non-aggregate Data Models
Graph Databases
Object Databases
Wide-column and Multi-model Databases
Aggregate Models vs. Non-aggregate Models
Advanced Querying and Scalability
Case Studies and Applications-1
Case Studies and Applications-2
Future Trends and Summary

Review Questions

Sl.No. - Questions	TLOs	BL
1. A retail store wants to store product information, where each product has a unique ProductID as a key and its details (like ProductName, Price, and Stock) as values. **Question**: Explain how a Key-Value database can efficiently store and retrieve this information compared to traditional relational databases.	1	2
2. An online bookstore uses a document database to store details of books, including attributes like Title, Author, Price, and Reviews. **Question**: Summarize how storing book details as documents provides flexibility in handling fields like Reviews, which can vary in number and content across books.	1	2
3. A social media application uses a column-family store to store user profiles, where each profile contains variable attributes like Name, Posts, Likes, and Comments. **Question**: Describe how column-family databases store this data efficiently and allow queries based on specific columns like Likes.	1	2
4. A research team uses a graph database to store relationships between authors and their published papers. **Question**: Explain how non-aggregate data models are better suited for representing complex relationships compared to aggregate data models.	1	2
5. A grocery delivery app needs to manage a large volume of key-value pairs, where the key is the OrderID and the value includes details like CustomerName, Items, and TotalCost. **Question**: Design a use case showing how to store, retrieve, and update an order in a Key-Value database when a customer modifies their order.	2	3
6. A movie streaming service wants to store details of movies in a document database. Each document includes fields like Title, Genre, Cast, and Ratings. **Question**: Demonstrate how to design and query a document database to retrieve movies in the "Action" genre with ratings above 4.5.	2	3
7. A telecom company stores call records in a column-family database, with columns for CallerID, ReceiverID, CallDuration, and Date. **Question**: Apply your knowledge of column-family stores to create a schema for this use case and write a query to retrieve all calls made by a specific CallerID in the last week.	2	3

8. A logistics company uses a graph database to track packages. Nodes represent Locations, and edges represent Routes between them, with weights for Distance. **Question**: Design a query to find the shortest route between two locations using a graph-based non-aggregate data model.	3	3

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 4. Implementation of aggregates	Planned Hours: **12 hrs**

Learning Outcomes :-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Apply the selection of an aggregate-based database for a given scenario.	2	3	1
2. Demonstrate aggregate data models.	2	3	2
3. Compare the strengths and limitations of data modeling in Riak, MongoDB, and Cassandra.	3	3	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to Aggregates and Aggregate Databases
2. Aggregates in Riak
3. Aggregates in MongoDB
4. Aggregates in Cassandra
5. Comparative Analysis of Aggregate Implementations
6. Comparative Analysis of Aggregate Implementations
7. **Modeling Data for Access** Principles of Data Modeling
8. Modeling Data in Riak

9. Modeling Data in MongoDB
10. Modeling Data in Cassandra
11 Comparative Analysis and Case Studies
12. Comparative Analysis and Case Studies

Review Questions

Sl.No. - Questions	TLOs	BL
1. A retail company wants to use Riak to handle its distributed inventory system where data is replicated across multiple nodes for high availability. **Question**: Explain how Riak ensures data availability and fault tolerance in this distributed system.	1	2
2. An online bookstore uses MongoDB to store book details such as title, author, category, and price. Each book can have multiple reviews. **Question**: Illustrate how MongoDB's document-based structure would represent this data.	1	2
3. A social media platform uses Cassandra to store billions of user interactions (likes, shares, and comments) per day. **Question**: Describe how Cassandra's partitioning and replication mechanisms make it suitable for handling this massive workload.	1	2
4. A logistics company needs to design its database to ensure fast retrieval of shipment tracking details. **Question**: Explain why modeling data based on access patterns is crucial and provide an example of a suitable data structure.	1	2
5. A banking application wants to manage data aggregates, such as customer accounts and transactions, efficiently. **Question**: Explain the concept of aggregates and how it helps in organizing data for consistency and performance in distributed databases.	1	2
6. A video streaming service uses Riak to store user preferences and playback history. Each user's data is replicated across three nodes. **Question**: Demonstrate how Riak would retrieve a user's playback history if one of the nodes fails.	2	3
7. A fitness app stores user profiles in MongoDB, including details like name, age, and a list of daily activities. **Question**: Write a MongoDB query to insert a new user's profile and retrieve all users who performed "running" as an activity.	2	3

8. An IoT system records sensor readings (timestamp, temperature, and humidity) from multiple devices into Cassandra. **Question**: Demonstrate how to design a table in Cassandra to efficiently query the temperature readings for a specific device within a given date range.	2	3
9. A food delivery service needs to model its data for fast access to orders based on delivery location and time. **Question**: Design a data model that supports these access patterns and explain why your model is efficient.	2	3
10. A gaming platform wants to aggregate player statistics (e.g., total score, games played, and rank) for quick leaderboard updates. **Question**: Demonstrate how to model and store these aggregates in a distributed database like MongoDB or Cassandra to ensure real-time leaderboard updates.	2	3

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 5. Capturing relations and At Scale	Planned Hours: **13 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the concept of relationships in data and how they are represented.	3	2	1
2. Demonstrate relational data models for specific scenarios.	3	3	2
3. Compare use of the relational data models in distributed environment.	3	2	1

Lesson Schedule Class No. - Portion covered per hour / per Class
Introduction to Aggregates
Querying and Analyzing Aggregates
Impact of Aggregates on Database Design
Capturing Relationships in Data
Challenges in Capturing Relations
Data with Relations
Constructing Relational Models
Introduction to Graph Databases
Neo4j Basics
Distribution Models
Consistency and Version Stamps
Introduction to Map-Reduce
Deployment and Cluster Configuration

Review Questions

Sl.No. - Questions	TLOs	BL
1. A social networking platform wants to represent users, their friendships, and the posts they share. **Question**: Explain how a graph database like Neo4j can be used to model this scenario, including nodes, relationships, and properties.	1	2
2. A company uses a relational database to store employee data and their reporting structure. **Question**: Compare how storing hierarchical relationships (e.g., manager-employee reporting) differs between relational databases and graph databases.	3	2
3. An e-commerce website wants to recommend products based on customer purchase patterns. The data includes customers, products, and the relationships between them (e.g., "bought," "viewed"). **Question**: Using Neo4j, design a graph model to represent this data. Write a Cypher query to find customers who bought a specific product and recommend related products based on other customers' purchases.	2	3
4. A transport company wants to model a city's metro system, where stations are nodes and connections between them are relationships with	2	3

properties like distance and travel time. **Question**: Demonstrate how to create this model in Neo4j and write a Cypher query to find the shortest path between two stations.		
5. A multinational organization uses a distributed database to manage its data across regions. **Question**: Explain how distribution models impact data availability and consistency, and provide an example of when eventual consistency is preferred over strong consistency.	1	2
6. A company needs to process large-scale data (e.g., analyzing customer behavior across millions of transactions). **Question**: Describe how the Map-Reduce framework works and explain why it is suitable for this scenario.	3	2
7. Scenario: A cloud-based service processes user data stored across distributed servers to ensure high availability and fault tolerance. **Question**: Apply your knowledge of distribution models to propose a deployment strategy that ensures minimal downtime during server failures. Include the use of version stamps for data synchronization.	2	3
8. A marketing agency collects billions of ad-click logs daily. They need to calculate the total clicks per region efficiently. Question: Write and explain the steps of a Map-Reduce job to process the ad-click data and calculate the required totals.	2	3

Chapterwise Plan

Course Code and Title: SC2113 - Advanced Database Concepts	
Chapter Number and Title: 6. As system	Planned Hours: **10 hrs**

Learning Outcomes:-

At the end of the topic the student should be able to:

Topic Learning Outcomes	COs	BL	PO No
1. Explain the advanced concepts like schema migration, polyglot persistence, and multi-model databases.	4	2	1
2. Demonstration of advanced concepts.	4	3	2

3. Choosing appropriate database for given application	4	3	1

Lesson Schedule Class No. - Portion covered per hour / per Class
1. Introduction to Schema Migration
2. Techniques for Schema Migration
3. Introduction to Polyglot Persistence
4. Case Studies in Polyglot Persistence
5. Multi-model Database Types
6. Comparison of Multi-model and Specialized Databases
7. Beyond NoSQL
8. Choosing the Right Database
9. Practical Implementation and Integration
10. Wrap-Up and Case Study Discussion

Review Questions

Sl.No. - Questions	TLOs	BL
1. A company is moving its database from MySQL to PostgreSQL to leverage better scalability and advanced features. **Question**: Explain the process of schema migration in this context. How does it ensure data integrity during the migration?	1	2
2. A retail company uses a relational database for storing transaction data and a NoSQL database for managing customer reviews and unstructured data. **Question**: Describe how polyglot persistence allows the company to optimize its data storage and retrieval. Why is it preferred over using a single database system?	1	2
3. A social media platform stores user data as a graph, posts as documents, and transactional data in a relational database. **Question**: Explain how a multi-model database type can integrate these data models effectively. What are the advantages of this approach?	1	2
4. A financial institution is evaluating databases that combine relational and NoSQL features, such as NewSQL databases. **Question**: Discuss how NewSQL databases address the limitations of traditional relational and NoSQL systems.	1	2

5. A startup needs a database for real-time analytics, high write throughput, and scalability. **Question**: Explain the factors they should consider when choosing the right database. Provide examples of suitable database systems for this use case.	1	2
6. An e-commerce company is migrating its schema from a NoSQL database to a relational database to meet reporting requirements. **Question**: Demonstrate the steps involved in migrating the schema while maintaining data consistency. Include an example of migrating a "Products" collection with nested documents into relational tables.	2	3
7. A travel booking platform uses a relational database for transaction records, a key-value store for session management, and a document database for user preferences. **Question**: Design a data flow showing how these databases interact for booking a flight. Write a query or command for each database type to demonstrate this interaction.	2	3
8. A content delivery platform stores metadata for videos in a graph database, user reviews in a document database, and subscription details in a relational database. **Question**: Demonstrate how to retrieve data about a video, including its metadata, reviews, and subscription requirements, by integrating these data models.A healthcare system needs a database for handling massive patient data with ACID properties, real-time analytics, and scalability. **Question**: Demonstrate how a NewSQL database (e.g., Google Spanner or CockroachDB) can address these requirements. Include an example of schema design and query.	2	3
9. A gaming application requires low latency for in-game data, high throughput for event tracking, and schema flexibility for storing user-generated content. **Question**: Apply your knowledge to recommend a database solution, justify your choice, and design a basic schema for one of the required use cases (e.g., event tracking).	3	3

MIT VISHWAPRAYAG UNIVERSITY

School of Computing

Model QP Minor 1

A.Y.2025-26 – Sem – I

Program :

Course Title: Advanced Database Concepts	Course Code: SC2113
Max Marks : 30	**Duration : 1 hrs**

Instructions: The question paper consists of 3 questions, each carrying 15 marks. Answer any 2 questions.

Q. No	Question	M	BLL	CO
Q1	a) Define the key characteristics of RDBMS. How is it different from a traditional file-processing system?	5	**2**	**1**
	b) Explain an ER diagram for an e-commerce platform (entities: Customers, Orders, Products). Specify attributes, primary keys, and relationships.	5	**2**	**1**
	c) Explain Relational Algebra. Write a query to retrieve the names of customers who placed orders worth more than ₹10,000.	5	**2**	**1**
Q2	a) Compare and contrast Key-Value Databases, Document Databases, and Column-Family Stores in terms of data storage and use cases.	5	**2**	**2**
	b) Explain how data for an online library system can be stored in Key-Value, Document, and Column-Family databases. Describe their differences.	5	**2**	**2**
	c) Why are aggregate-based data models more suitable for large-scale distributed systems than relational databases?	5	**2**	**2**
Q3	a) What are Non-Aggregate Data Models? Provide examples of databases that use these models and explain their key features.	5	**2**	**2**

	b) Discuss scenarios where migrating from a relational database to a NoSQL database is beneficial. Highlight challenges during such migrations.	5	**2**	**2**
	c) Explain Polyglot Persistence in modern database systems. Provide an example where multiple database models are used together in a real-world application.	5	**2**	**2**

MIT VISHWAPRAYAG UNIVERSITY

School of Computing

Model QP Minor 2

A.Y.2025-26 – Sem – I

Program :

Course Title: Advanced Database Concepts	Course Code: SC2113
Max Marks : 30	**Duration : 1 hrs**

Instructions: The question paper consists of 3 questions, each carrying 15 marks. Answer any 2 questions.

Q. No	Question	Marks	BLL	CO
Q1	**a)** Explain the concept of aggregate-based data models with examples of **Riak**, **MongoDB**, and **Cassandra**. Highlight their use cases in large-scale applications.	5	**2**	**3**
	b) Design a data model for an e-commerce application in **MongoDB**, where you need to store details about Products and Orders. Explain how the data is structured for efficient access.	5	**2**	**3**
	c) Compare and contrast **Relational Databases** with **Graph Databases**. How does **Neo4j** handle data with complex relationships effectively?	5	**2**	**3**
Q2	**a)** What are the main challenges in distributing data across clusters in large-scale systems? Explain the **Distribution Models** used in databases like Cassandra.	5	**2**	**3**
	b) Discuss the concept of **Consistency** in distributed databases. Compare eventual consistency with strong consistency using examples.	5	**2**	**3**

	c) Explain how **Map-Reduce** works as a data processing technique. Illustrate with an example of analyzing large-scale log files in a distributed environment.	5	**2**	**3**
Q3	**a)** Write the steps for **cluster configuration** and deployment of a distributed database system. Highlight key considerations in setting up a database like **Cassandra** or **Riak**.	5	**2**	**3**
	b) A social media platform needs to model user interactions, such as friendships and likes, using a **Graph Database**. Explain how you would design this data model in **Neo4j**.	5	**2**	**3**
	c) What are **Version Stamps** in distributed databases? Discuss their role in ensuring consistency in large-scale systems.	5	**2**	**3**

MIT VISHWAPRAYAG UNIVERSITY
School of Computing
Model Question Paper End Semester Examination
A.Y.2025-26 – Sem – I

Program: BCA **Date:**
Course Code:SC2113 **Course:** Advanced Database Concepts
Max Marks: 100 **Duration: 3 Hrs**

Section A -UNIT I

QN		Question (Answer ANY TWO full questions from Q.1, Q.2 & Q.3)	Level	Marks	CO
1	**A**	1. You have been tasked with designing a database for a library system. The system needs to store information about books, authors, and borrowers. Based on this scenario, draw an Entity-Relationship (ER) diagram and explain the entities, relationships, and attributes involved.	2	4	1
		2. Draw and explain an Entity-Relationship (ER) diagram for a simple library system.	2	4	1
	B	1. The library database needs to be designed using a Relational Database Management System (RDBMS). Based on the scenario, explain the steps you would follow to design the relational schema for the system.	3	6	1
		2. The library system has started facing issues like handling unstructured data such as scanned images of books or PDFs in the catalog. Discuss the shortcomings of relational databases in managing such unstructured data and suggest an alternative data model.	3	6	1
2	**A**	1. A startup e-commerce company wants to implement a simple data storage system to track customer orders. They decide to use a Key-Value database for scalability and flexibility. Define Key-Value Databases and explain how they are suitable for this use case.	2	8	2
	B	1. The e-commerce company later decides to implement a Document Database to store customer profiles along with their order history in a flexible format. Discuss how Document Databases would work for this case and describe the architecture.	3	6	2
		2. The company is now facing challenges with managing product information and inventory across multiple locations. They are considering Column-Family stores for the task.	3	6	2

		Explain why Column-Family stores would be more efficient than Key-Value or Document Databases in handling this scenario.			
3	**A**	A social media platform wants to store user data, such as profiles, posts, and comments, where the data structure may evolve over time. They are considering a Document Database for flexibility in handling semi-structured data. Define Document Databases and explain how they can address the needs of the social media platform.	2	8	2
	B	1. The social media platform wants to implement a new feature where users can search for posts containing certain keywords and tags. The system needs to be scalable to handle millions of users and posts. Discuss the advantages of using a Document Database for this feature and describe how data is indexed and queried. (6 Marks)	3	6	2
		2. After evaluating other storage options, the platform is considering using a Column-Family store to improve the performance of the new feature. Explain the advantages and potential challenges of using Column-Family stores for such large-scale data operations. (6 Marks)	3	6	2
Section B -UNIT II					
QN		**Question (Answer ANY TWO full questions from Q.4, Q.5 & Q.6)**	**Level**	**Marks**	**CO**
4	**A**	Define relational databases and explain how aggregate functions are implemented in these systems. Give examples of common aggregate operations like COUNT, SUM, AVG, MAX, and MIN.	2	8	3
	B	1. Compare and contrast the implementations of aggregates in the following NoSQL databases: Riak, MongoDB, and Cassandra. Discuss the strengths and weaknesses of each in handling large-scale data aggregation.	3	6	3
		2. A company is using MongoDB to store product information and needs to generate reports for sales figures aggregated by product category. The dataset is expected to grow rapidly. How would you implement the aggregation logic in MongoDB to ensure performance and scalability? Discuss any trade-offs or limitations you might encounter in this approach.	3	6	3
5	**A**	Explain the concept of capturing relations in a database. What are the key differences between relational databases and graph databases when it comes to modeling relationships?	2	8	3
	B	1. Discuss the use of Neo4j as a graph database for capturing relations. Provide an example of how you would model a social	3	6	3

		network in Neo4j, including the nodes, relationships, and properties involved.			
		2. A company is building a recommendation system for its e-commerce platform using Neo4j to model product recommendations based on user behavior and preferences. How would you model the relationships between users, products, and recommendations in Neo4j? What queries would you use to recommend products to users based on their interaction history?	3	8	3
6	**A**	What is a distribution model in the context of databases? Explain the different types of distribution models commonly used in NoSQL systems, such as sharding and replication.	2	8	3
	B	1. In a distributed database system, explain the concepts of consistency, version stamps, and the role they play in ensuring data integrity across nodes. How do these concepts relate to the CAP theorem?	3	6	3
		2. You are working with a distributed database system that uses sharding and replication for data storage. The system has experienced a temporary network failure between some of its nodes. What consistency issues might arise during the failure, and how would you resolve them to ensure data accuracy and reliability once the network is restored?	3	6	3
Section C -UNIT III					
QN		**Question (Answer ANY ONE full question from Q.7 & Q.8)**	**Level**	**Marks**	**CO**
7	**A**	Explain the concept of **schema migration** in database systems. Why is it necessary in the context of evolving database schemas?	2	8	4
	B	1. A company is moving from a traditional relational database (SQL) to a NoSQL database for improved scalability and flexibility. The existing schema includes tables with rigid relationships and a lot of foreign key constraints. • What are the key challenges that might arise during schema migration? • How would you handle these challenges while migrating to a NoSQL database?	3	6	4
		2. Consider a large e-commerce platform that needs to store various types of data, such as user profiles, orders, product catalogs, and reviews. Some data is structured, some semi-structured, and some unstructured. • Given this scenario, which type of **polyglot persistence** approach would you recommend? Explain why it's a suitable choice for the platform.	3	6	4

8	**A**	Describe **multi-model databases** and explain how they differ from traditional relational databases. Provide examples of use cases where multi-model databases can be advantageous.	2	8	4
	B	1. A startup is developing a social media platform that needs to manage various data models such as user profiles, relationships between users, messages, and posts. They are considering using a **multi-model database** to handle these requirements. • What are the benefits of using a multi-model database in this scenario? How would it help the platform to scale and remain flexible?	3	6	4
		2. Your company is considering implementing a NoSQL solution for a real-time analytics system where performance, scalability, and data flexibility are key priorities. **Beyond NoSQL**, what other databases or architectures might be considered? Explain why you would recommend them based on the requirements of real-time analytics.	3	6	4

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Course Plan

Semester: 04 Year: 2025-26

Course Title: **Full Stack Development II**	Course Code: **SC2037**
L-T-P : 0-0-2	Credits: 02
Total Contact hrs: 60	Duration of ESA: 3 hours
ISA Weightage: 80	ESA Weightage:20
Lesson Plan Authors: Dr. Savita Hanji	Date:

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Apply node.js and express.js fundamentals to build applications.
2. Apply middlewares in express applications
3. Create restful apis with express
4. Implement authentication and authorization
5. Deploy node.js applications to production environments

Course Articulation Matrix: Mapping of Course Outcomes (CO) with Program Outcomes

Course Title: Full Stack Development II	Semester:4
Course Code:SC2037	Year: 2025-2026

Course Outcomes / Program Outcomes	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Apply node.js and express.js fundamentals to build applications.	3													
Apply middlewares in express applications	3													
Create restful apis with express	3													
Implement authentication and authorization	3													

Deploy node.js applications to production environments	3													

Experiment wise Plan

List of experiments/jobs planned to meet the requirements of the course.

Category: Exercise		
Expt./ Job No.	**Experiment / Job Details**	**Marks / Experiment**
1	Build a command-line task manager application that allows the user to add, view, edit, and delete tasks. Store task data in a JSON file and implement asynchronous operations using Node.js. Key Features: i. Create tasks with a title and description. ii. List all tasks stored in the JSON file. iii. Update a task's status to "completed." iv. Delete a task. v. Save and load tasks from a JSON file.	5.00
2	Create an Express application that handles customer feedback. The app should allow customers to submit feedback via a form. The feedback will be stored in a simple in-memory structure, and an administrator should be able to view, edit, or delete feedback. Key Features: i. A customer-facing feedback form for input. ii. Admin routes for viewing all feedback submissions. iii. Update and delete feedback functionalities for admin.	5.00

3	Develop a RESTful API for managing employee information. The API will allow the creation, retrieval, updating, and deletion of employee records. Key Features: i. Implement full CRUD operations for employee management (Create, Read, Update, Delete). ii. API should return JSON responses and handle validation errors.	5.00
4	Build a task management system that stores tasks in MongoDB. Use Mongoose to define schemas and interact with the database. Key Features: i. Add tasks with details like task name, priority, and due date. ii. Update task details and mark tasks as completed. iii. Delete tasks when they are no longer needed. iv. Use pagination and search filters to find tasks by priority or due date.	5.00
5	Build a secure authentication system using JWT for login, registration, and token validation. Key Features: i. Secure user registration and login. ii. JWT-based token authentication. iii. Token validation middleware.	5.00
6	Build a RESTful API with authentication, task creation, updating, and deletion, all using MongoDB. Key Features: i. API with JWT authentication. ii. Full CRUD functionality for tasks. iii. Handling asynchronous database operations.	5.00

7	Write unit and integration tests for the To-Do list application built in previous topics. Key Features: i. Unit tests for task operations. ii. Integration tests for the full API.	5.00
8	Deploy the completed To-Do List API to Heroku and set up continuous integration (CI) using GitHub Actions. Key Features: i. Deploying to Heroku with MongoDB integration. ii. Setting up environment variables and database connections.	5.00
9	Integrate the Node.js Task Manager API with a frontend built using HTML, CSS, and JavaScript (Vanilla or React). Key Features: i. User interface for creating and managing tasks. ii. Integration with the backend API.	5.00

Experiment / Job Plan

Laboratory Title:**Full Stack Development II**	
Expt./Job No.:**1.** Build a command-line task manager application that allows the user to add, view, edit, and delete tasks. Store task data in a JSON file and implement asynchronous operations using Node.js. Key Features: Create tasks with a title and description. List all tasks stored in the JSON file. Update a task's status to "completed." Delete a task. Save and load tasks from a JSON file.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Explain Node.js fundamentals and set up Node.js and npm environment.	CO1	L2
2	Work with Node.js modules and perform asynchronous operations.	CO1	L3
3	Explain and implement the difference between CommonJS and ES Modules in Node.js.	CO1	L3

Expt./Job No.:**2.** Create an Express application that handles customer feedback. The app should allow customers to submit feedback via a form. The feedback will be stored in a simple in-memory structure, and an administrator should	Category: Exercise

be able to view, edit, or delete feedback. Key Features: A customer-facing feedback form for input. Admin routes for viewing all feedback submissions. Update and delete feedback functionalities for admin.	

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Set up and configure an Express.js application.	CO2	L3
2	Implement basic routes and HTTP methods (GET, POST).	CO2	L3
3	Explain the structure and flow of an Express application.	CO1	L2

Expt./Job No.:**3.** Develop a RESTful API for managing employee information. The API will allow the creation, retrieval, updating, and deletion of employee records. Key Features: Implement full CRUD operations for employee management (Create, Read, Update, Delete). API should return JSON responses and handle validation errors.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Set up routes to handle CRUD operations in an Express app.	CO3	L3
2	Implement HTTP methods (GET, POST, PUT, DELETE)	CO3	L3
3	Handle responses and request data in RESTful APIs.	CO3	L3

Expt./Job No.:**4.** Build a task management system that stores tasks in MongoDB. Use Mongoose to define schemas and interact with the database. Key Features: Add tasks with details like task name, priority, and due date. Update task details and mark tasks as completed. Delete tasks when they are no longer needed. Use pagination and search filters to find tasks by priority or due date.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Set up and connect to MongoDB using Mongoose.	CO3	L3
2	Define Mongoose schemas for data models.	CO3	L3

3	Perform CRUD operations using Mongoose methods.	CO3	L3

Expt./Job No.:**5.** Build a secure authentication system using JWT for login, registration, and token validation. Key Features: i. Secure user registration and login. J ii. WT-based token authentication. iii. Token validation middleware.	Category**:** Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Implement JWT authentication in a Node.js app.	CO4	L3
2	Understand and implement Passport.js for user authentication.	CO4	L3
3	Secure routes using JWT authentication middleware.	CO4	L3

Expt./Job No.:6. Build a RESTful API with authentication, task creation, updating, and deletion, all using MongoDB. Key Features: i. API with JWT authentication. ii. Full CRUD functionality for tasks.	Category: Exercise

iii. Handling asynchronous database operations.	

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Design and implement a RESTful API with Express and MongoDB.	CO3	L3
2	Implement user authentication for secured API access.	CO4	L3
3	Handle asynchronous database operations with MongoDB and Mongoose.	CO3	L3

Expt./Job No.:7. Write unit and integration tests for the To-Do list application built in previous topics. Key Features: i. Unit tests for task operations. ii. Integration tests for the full API.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Set up Mocha and Chai for unit testing in Node.js.	CO1	L3
2	Write unit tests for individual application functions.	CO1	L3
3	Perform integration testing for API endpoints.	CO1	L3

Expt./Job No.:8. Deploy the completed To-Do List API to Heroku and set up continuous integration (CI) using GitHub Actions. Key Features: i. Deploying to Heroku with MongoDB integration. ii. Setting up environment variables and database connections.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Deploy a Node.js application to Heroku.	CO5	L3

2	Configure and use environment variables in production.	CO5	L3
3	Set up continuous integration and deployment (CI/CD) workflows.	CO5	L3

Expt./Job No.: 9. Integrate the Node.js Task Manager API with a frontend built using HTML, CSS, and JavaScript (Vanilla or React). Key Features: i. User interface for creating and managing tasks. ii. Integration with the backend API.	Category: Exercise

Learning Outcomes:-

At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Set up a basic frontend project with HTML, CSS, and JavaScript or React.	CO1	L3
2	Integrate frontend with a RESTful API using AJAX or fetch or axios.	CO1	L3
3	Implement basic front-end validation and error handling.	CO1	L3

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Course Plan

Laboratory Title: **Exploratory Data Analysis and Data Visualization**	Lab. Code: **SC2301**
Total Hours: **60**	Duration of SEE Hours: 4
SEE Marks: **20**	CIE Marks: **80**
Lab. Plan Author: Mr.Krishnendu Jana	Date:
Checked By: Mrs.Savita Hanji	Date:

Course Outcomes (COs):

At the end of the course the student should be able to:

1. Explain the need for and the basic concepts of exploratory data analysis (eda).
2. Apply appropriate eda techniques to real-world data.
3. Analyze eda output data to draw meaningful inferences.
4. Create effective data visualizations to communicate and report results clearly.

Course Articulation Matrix: Mapping of Course Outcomes (CO) with Program Outcomes

Course Title:Exploratory Data Analysis and Data Visualization	Semester:4
Course Code:SC2301	Year:

Course Outcomes / Program Outcomes	1	2	3	4	5	6	7	8	9	10	11	12
Explain the need for and the basic concepts of exploratory data analysis (eda).	3											
Apply appropriate eda techniques to real-world data.					3							
Analyze eda output data to draw meaningful inferences.				3								
Create effective data visualizations to communicate and report results clearly.				3								

Text Books (List of books as mentioned in the approved syllabus)

1. Brent Dykes, Effective Data Storytelling, 1st, Wiley, 2019
2. Claus O. Wilke, Fundamentals of Data Visualization, 1st, O'Reilly Media, 2019
3. Allen B. Downey, Think Stats: Exploratory Data Analysis in Python, 2nd, O'Reilly Media, 2014

References

1. Wes McKinney, Python for Data Analysis, 3rd, O'Reilly Media, 2022

Experiment wise Plan

List of experiments/jobs planned to meet the requirements of the course.

Category: Demonstration		**No. of lab sessions: 60.00**		
Expt./ Job No.	**Experiment / Job Details**	**Marks / Experiment**	**Planned Delivery date**	**Actual Delivery date**
1	Foundations of EDA and Python	10.00		
2	Data Manipulation with Pandas	10.00		
3	Data Collection and Cleaning	10.00		
4	Descriptive Statistics and Variate Analysis	15.00		
5	Advanced EDA and Dimensionality Reduction	15.00		
6	Data Visualization Principles and Project	20.00		

Experiment / Job Plan

Laboratory Title:**Exploratory Data Analysis and Data Visualization**	
Expt./Job No.:**1. Foundations of EDA and Python**	Category:**Demonstration**

Learning Outcomes:- At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Explain the purpose of Exploratory Data Analysis and describe the key stages of the EDA process.	CO1	L2
2	Apply fundamental Python programming constructs, including loops, functions, and libraries like NumPy and Pandas, to load and manipulate data structures.	CO2	L3

Lesson Schedule:-

Lecture No. - Portion covered per hour
1. Definition and importance of EDA, common challenges, and the EDA process.
2. L - Introduction to Python environment (Jupyter Notebook/Google Colab). Basic Python syntax, variables, and data types.
3. Loops (for, while), conditional statements (if, else).
4. L - Writing simple Python scripts involving loops and conditionals to solve basic problems.
5. Defining and calling functions, string manipulation and regular expressions.
6. L - Creating custom functions for data manipulation tasks and practicing string operations with real-world examples.
7. Introduction to NumPy, arrays, and basic array operations.

8. L - Creating and manipulating NumPy arrays, performing mathematical operations on arrays.

Review/Assignment Questions:-

Sl.No. - Questions
1. Write a python program for the below problem Given a string `s` consisting of words and spaces, return *the length of the **last** word in the string.* A **word** is a maximal substring consisting of non-space characters only. **Example 1:** **Input:** s = "Hello World" **Output:** 5 **Explanation:** The last word is "World" with length 5. **Example 2:** **Input:** s = " fly me to the moon " **Output:** 4 **Explanation:** The last word is "moon" with length 4.
2. Write a python program for the below problem Given two strings `needle` and `haystack`, return the index of the first occurrence of `needle` in `haystack`, or `-1` if `needle` is not part of `haystack`. **Example 1:** **Input:** haystack = "sadbutsad", needle = "sad" **Output:** 0 **Explanation:** "sad" occurs at index 0 and 6. The first occurrence is at index 0, so we return 0. **Example 2:** **Input:** haystack = "leetcode", needle = "leeto" **Output:** -1 **Explanation:** "leeto" did not occur in "leetcode", so we return -1.

3.

Write a Python function called `parse_log_entry` that takes a single log entry string as input. The log entry is semi-structured and contains a timestamp, a log level, and a message. Your function should use regular expressions to extract these three components and return them in a dictionary with the keys 'timestamp', 'level', and 'message'. If a component is not found, its value in the dictionary should be `None`.

Test Cases:

- **Input:** `"2024-07-21 10:55:01,INFO,User 'alice' logged in successfully."`
  Expected Output:
  `{'timestamp': '2024-07-21 10:55:01', 'level': 'INFO', 'message': "User 'alice' logged in successfully."}`
- **Input:** `"2024-07-21 11:00:05,ERROR,Failed to process payment for transaction 9876."`
  Expected Output:
  `{'timestamp': '2024-07-21 11:00:05', 'level': 'ERROR', 'message': 'Failed to process payment for transaction 9876.'}`
- **Input:** `"WARNING,System memory usage is high."`
  Expected Output:
  `{'timestamp': None, 'level': 'WARNING', 'message': 'System memory usage is high.'}`
- **Input:** `"2024-07-21 11:15:00,DEBUG,"`
  Expected Output: `{'timestamp': '2024-07-21 11:15:00', 'level': 'DEBUG', 'message': None}`

4.

Create a Python function named `clean_and_format_product_codes` that takes a list of raw product codes (strings). This function should perform the following operations for each product code:

1. Remove any leading or trailing whitespace.
2. Convert the entire string to uppercase.
3. Replace any hyphens (-) or underscores (_) with an empty string.
   The function should return a new list containing the cleaned and formatted product codes.

Test Cases:

- **Input:** `[' prod-1234 ', 'item_5678', ' PART-91011 ']`
  Expected Output: `['PROD1234', 'ITEM5678', 'PART91011']`
- **Input:** `[' SKU_ABC-XYZ ', ' DEF-456_GHI ']`
  Expected Output: `['SKUABCXYZ', 'DEF456GHI']`
- **Input:** `['no-special-chars', 'ALLUPPER']`
  Expected Output: `['NOSPECIALCHARS', 'ALLUPPER']`
- **Input:** `[]`
  Expected Output: `[]`

5. Write a Python function called `analyze_sensor_data` that takes a NumPy array of temperature readings. The function should perform the following calculations and return the results in a dictionary:

- The average temperature.
- The highest and lowest temperatures.
- The number of readings above a certain threshold (e.g., 25.0 degrees).
  The threshold should be an optional parameter to the function.

Test Cases:

- **Input:** `readings = np.array([22.5, 24.1, 26.3, 21.9, 28.0, 25.5]), threshold = 25.0`
  Expected Output:
  `{'average_temp': 24.7166..., 'max_temp': 28.0, 'min_temp': 21.9, 'readings_above_threshold': 3}`
- **Input:** `readings = np.array([15.0, 18.5, 12.3, 20.1]), threshold = 20.0`
  Expected Output:
  `{'average_temp': 16.475, 'max_temp': 20.1, 'min_temp': 12.3, 'readings_above_threshold': 1}`

Expt./Job No.:**2. Data Manipulation with Pandas**	Category:**Demonstration**

Learning Outcomes:- At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Apply a variety of Pandas functions to select, filter, group, and aggregate data to answer specific questions.	CO2	L3
2	Solve data integrity issues by implementing methods for handling missing values and removing duplicates from a real-world dataset.	CO2	L3

<u>Lesson Schedule:-</u>

Lecture No. - Portion covered per hour
1. Introduction to Pandas DataFrames and Series.

2. L - Loading datasets into Pandas DataFrames from different file formats (CSV, Excel).
3. Basic DataFrame operations: head(), tail(), info(), describe().
4. L - Inspecting a new dataset to understand its structure, data types, and basic statistics.
5. Indexing, slicing, and filtering data in Pandas.
6. L - Selecting specific rows and columns from a DataFrame based on conditions.
7. Handling missing values and duplicates.
8. L - Applying techniques to clean a dataset by dropping or filling missing values and removing duplicate entries.

Review/Assignment Questions:-

Sl.No. - Questions

1. Create a Python function named `load_dataset` that takes a file path as input. The function should check the file extension and load the file into a Pandas DataFrame. It should support both '.csv' and '.xlsx' (Excel) files. If the file path has a different extension or the file does not exist, the function should return `None`.
• **Setup for Test Cases:**
  ○ Create a file named `products.csv` with the content:
  `product_id,product_name,price`
  `1,Laptop,1200`
  `2,Mouse,25`
  `3,Keyboard,75`
  ○ Create a file named `customers.xlsx` with similar data using a library like `openpyxl`.
Test Cases:
• **Input:** `'products.csv'`
Expected Output: A Pandas DataFrame containing the data from the CSV file.
• **Input:** `'customers.xlsx'`
Expected Output: A Pandas DataFrame containing the data from the Excel file.
• **Input:** `'data.txt'`
Expected Output: None
• **Input:** `'non_existent_file.csv'`
Expected Output: None

2. Write a Python function called `get_dataframe_summary` that accepts a Pandas DataFrame as input. The function should return a dictionary containing:

1. The first 3 rows of the DataFrame (using `head(3)`).
2. The last 3 rows of the DataFrame (using `tail(3)`).
3. The summary statistics of the numerical columns (using `describe()`).

Test Cases:

- **Input:** A DataFrame `df` created as follows:

```
import pandas as pd
import numpy as np
data = {'A': np.arange(10), 'B': np.random.rand(10), 'C':
list('abcdefghij')}
df = pd.DataFrame(data)
```

Expected Output: A dictionary where the keys 'head', 'tail', and 'describe' map to the corresponding DataFrames from the operations. The exact values for 'B' will vary due to randomness, but the structure should be correct.

Problem Statement:

You are given a raw dataset of customer feedback entries as a multi-line string in CSV format. Write a single Python function called `process_and_filter_feedback` that takes this string as input and performs a full cleaning and filtering workflow.

The function must perform the following steps:

1. **Load Data**: Convert the input CSV string into a Pandas DataFrame. The columns will be `feedback_id`, `customer_id`, `rating`, and `comment`.
2. **Clean the Data**:
   - The `rating` column may contain non-numeric values; convert it to a numeric type, coercing any errors into NaN (Not a Number).
   - Remove any rows that are complete duplicates.
   - Drop any rows where `rating` or `comment` values are missing.
3. **Feature Engineering**:
   - Create a new boolean column named `is_negative`. A feedback entry is considered negative if its `rating` is less than or equal to 2.
4. **Filter and Select**:
   - From the cleaned data, filter for all "negative" feedback entries (`is_negative` is `True`).
   - From this filtered subset, select only the `customer_id` and `comment` columns.
5. **Return Value**: The function should return the final, filtered DataFrame containing only the `customer_id` and `comment` of negative feedback, sorted by `customer_id`.

3.

Test Case:

- **Input Data String:**

```
raw_data = """feedback_id,customer_id,rating,comment
1,101,1,The product broke after one use. This is urgent.
2,105,5,Excellent service!
3,102,2,The app is constantly crashing.
4,101,1,The product broke after one use. This is urgent.
5,103,4,It's okay, but could be better.
6,104,,"The rating was not recorded."
7,105,'invalid',This is not a real rating.
8,102,1,Very disappointed with the quality.
9,106,,This comment is missing a rating.
"""
```

- **Function Call:**

  Generated python

  ```
  process_and_filter_feedback(raw_data)
  ```

- **Expected Output (as a Pandas DataFrame):**

  ```
  customer_id                                comment
  0          101  The product broke after one use. This is urgent.
  1          102           The app is constantly crashing.
  2          102     Very disappointed with the quality.
  ```

  Explanation of the expected output:
  - Row with `feedback_id` 4 is a duplicate of row 1 and is removed.
  - Row 6 is dropped because `rating` is missing.
  - Row 7's `rating` ('invalid') becomes NaN during numeric conversion, so it's dropped.
  - Row 9 is dropped because its `comment` is missing a rating.
  - The remaining rows with ratings 1 and 2 are selected.
  - Finally, only the `customer_id` and `comment` columns are returned for these negative entries.

Expt./Job No.:**3. Data Collection and Cleaning**	Category:**Demonstration**

Learning Outcomes:- At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Apply appropriate techniques, such as web scraping or APIs, to collect data from structured and semi-structured sources.	CO2	L3
2	Analyze the impact of data transformation and outlier treatment on a dataset's distribution and summary statistics	CO3	L4

Lesson Schedule:-

Lecture No. - Portion covered per hour
1. Raw data, types of data (Structured, Unstructured, Semi-Structured), and sources.
2. L - Introduction to APIs and making simple API requests to fetch data.
3. Introduction to web scraping concepts and libraries (e.g., BeautifulSoup).
4. L - Scraping a simple webpage to extract structured data.
5. Outlier detection and handling techniques.
6. L - Identifying and treating outliers in a dataset using statistical methods.
7. Data transformation techniques (e.g., normalization, standardization).
8. L - Applying different data transformation methods and observing their impact on the data distribution.

Review/Assignment Questions:-

Sl.No. - Questions
1. You are a data analyst tasked with creating a unified dataset of product information for an e-commerce platform. The data is scattered across an internal API and a supplier's public webpage. Your goal is to write a single Python function, `create_processed_product_dataset`, that performs a full acquisition, cleaning, and transformation pipeline. The function will not take any arguments but will use pre-defined data sources (provided as variables below) to simulate the process. It must perform the following steps in order: 1. **Fetch from "API"**: You are given a dictionary, `api_data`, that mimics a JSON response from an internal API. It contains product IDs and prices. Load this into a Pandas DataFrame. 2. **Scrape from "Webpage"**: You are given an HTML string, `webpage_html`, containing a table with product details. Use a library like BeautifulSoup to scrape this table and load it into a second Pandas DataFrame. The columns should be `product_id`, `product_name`, and `stock_count`. 3. **Combine Data**: Merge the two DataFrames (from the API and the webpage) into a single master DataFrame based on `product_id`. 4. **Handle Outliers**: The `price` data is known to have entry errors. Identify outliers in the `price` column using the Interquartile Range (IQR) method. Any price above `Q3 + 1.5 * IQR` should be considered an outlier. Instead of removing these outliers, **cap** them by replacing their value with the calculated upper bound (`Q3 + 1.5 * IQR`). 5. **Transform Data (Standardization)**: To prepare the numerical data for a machine learning model, apply Z-score standardization to the (now capped) `price` column and the `stock_count` column. Store the results in new columns named `price_scaled` and `stock_scaled`. The formula for Z-score is: `(x - mean) / std_dev`. 6. **Return Value**: The function should return the final, processed DataFrame containing the merged data, the capped `price` column, and the new standardized columns.

Expt./Job No.:**4. Descriptive Statistics and Variate Analysis**	Category:**Demonstration**
Planned Delivery Date:	

Learning Outcomes:- At the end of the experiment/job the student should be able to:

Sl No.	Topic Learning Outcomes	COs	BL
1	Calculate measures of central tendency and dispersion to summarize data and apply univariate and bivariate analysis techniques.	CO2	L3
2	Analyze the output of correlation and bivariate analyses to derive relationships and patterns between different variables in a dataset.	CO3	L4

<u>Lesson Schedule:-</u>

Lecture No. - Portion covered per hour
1. Mean, median, and mode.
2. L - Calculating measures of central tendency for different variables in a dataset.
3. Variance, standard deviation, and range.
4. L - Calculating measures of dispersion to understand the spread of data.
5. Introduction to Matplotlib for basic plotting (histograms, box plots).
6. L - Creating histograms and box plots to visualize the distribution of single variables.
7. Introduction to Seaborn for enhanced visualizations.
8. L - Using Seaborn to create more aesthetically pleasing and informative univariate plots.
9. Scatter plots for numerical data.
10. L - Creating scatter plots to explore the relationship between two numerical variables.

11. Bar charts and grouped bar charts.
12. L - Visualizing the relationship between a categorical and a numerical variable using bar charts.
13. Pearson correlation coefficient and heatmaps.
14. L - Calculating and visualizing correlation matrices using heatmaps to identify relationships between variables.
15. Introduction to multivariate analysis concepts.
16. L - Exploring relationships between three or more variables using pair plots and 3D plots.

Review/Assignment Questions:-

Sl.No. - Questions	

1.	You are tasked with creating a function that performs a quick exploratory data analysis on a specific variable within a dataset. Write a Python function called `generate_variable_report` that takes a Pandas DataFrame and the name of a column (as a string) as input.
	
	The function should analyze the specified column and do the following:
	
	1. **Determine Data Type**: Check if the column is numerical (integer or float) or categorical (object/string).
	2. **For a Numerical Column**:
	  ○ Calculate the **mean**, **median**, **standard deviation**, and **range** (max - min).
	  ○ Generate two plots using Matplotlib or Seaborn:
	  ▪ A **histogram** to show the distribution of the data.
	  ▪ A **box plot** to identify the quartiles and potential outliers.
	  ○ Return a dictionary containing the calculated statistics (mean, median, std_dev, range). The function should also display the two plots.
	3. **For a Categorical Column**:
	  ○ Calculate the **mode** (most frequent category).
	  ○ Generate a **bar chart** (using Seaborn's `countplot`) showing the frequency of each category.
	  ○ Return a dictionary containing the calculated mode. The function should also display the plot.
	4. **Error Handling**: If the column name does not exist in the DataFrame, the function should raise a `ValueError` .

2. As a data analyst, you need to investigate how different variables in a dataset relate to each other. Write a Python function called `analyze_multivariate_relationships` that takes a Pandas DataFrame as input, along with the names of three columns: two numerical (`num_var1` , `num_var2`) and one categorical (`cat_var`).

The function should produce and display the following visualizations using the Seaborn library to explore the interactions between these variables:

1. **Correlation Heatmap**: Calculate the Pearson correlation matrix for all numerical columns in the entire DataFrame. Visualize this matrix as a **heatmap** with correlation values annotated on it.
2. **Grouped Visualization**: Create a visualization that shows the relationship between the three specified variables. A good choice is a **scatter plot** (`scatterplot`) where `num_var1` is on the x-axis, `num_var2` is on the y-axis, and the points are colored by the `cat_var` (`hue`).
3. **Comprehensive Overview**: Generate a **pair plot** (`pairplot`) for the entire DataFrame, which creates scatter plots for every pair of numerical variables and histograms for each individual numerical variable. The plots should also be colored by the `cat_var` .

The function does not need to return any data; its primary purpose is to generate and display these three distinct and informative plots.

Expt./Job No.:**5. Advanced EDA and Dimensionality Reduction**	Category:**Demonstration**
Planned Delivery Date:	

Learning Outcomes:-At the end of the experiment/job the student should be able to:

Sl No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Apply Principal Component Analysis (PCA) and clustering algorithms to reduce data dimensionality and segment data.	CO2	L3
2	Classify data points into distinct groups by analyzing the output of clustering algorithms and interpreting the resulting principal components.	CO3	L4

Lesson Schedule:-

Lecture No. - Portion covered per hour
1. Theory and application of Principal Component Analysis (PCA).
2. L - Applying PCA to a high-dimensional dataset to reduce the number of features.
3. Understanding principal components and explained variance.
4. L - Analyzing the results of PCA and selecting the optimal number of components.
5. Introduction to clustering algorithms (e.g., K-Means).
6. L - Applying K-Means clustering to segment data into distinct groups.
7. Metrics for evaluating clustering results (e.g., silhouette score).
8. L - Assessing the quality of clustering and interpreting the resulting clusters.

Review/Assignment Questions:-

Sl.No. - Questions

You are a data scientist at a retail company and have been given a high-dimensional dataset representing customer purchasing habits. Your goal is to segment these customers into distinct groups for targeted marketing campaigns. However, running clustering algorithms directly on high-dimensional data can be inefficient and less effective.
 	
Write a Python function called `run_customer_segmentation_pipeline` that takes a Pandas DataFrame of customer data and a desired number of clusters (`n_clusters`) as input. The function must perform an end-to-end analysis by following these steps:
 	
1. **Data Preparation**:
  o Isolate the numerical feature columns from the input DataFrame.
  o Standardize these features using a `StandardScaler` to ensure they have a mean of 0 and a standard deviation of 1. This is crucial for both PCA and K-Means.
2. **Dimensionality Reduction with PCA**:
  o Apply Principal Component Analysis (PCA) to the standardized data.
  o **Analyze Explained Variance**: Determine the minimum number of principal components required to explain at least **95%** of the total variance in the data. This number represents the "optimal" dimensionality for this dataset.
  o Transform the standardized data into this optimal number of principal components.
3. **Clustering with K-Means**:
  o Apply the K-Means clustering algorithm to the new, lower-dimensional data (the principal components) to segment the customers into `n_clusters`.
4. **Evaluation and Interpretation**:
  o Calculate the **silhouette score** of the clustering results to measure how well-defined the clusters are. A score closer to 1 indicates better-defined clusters.
  o Add the resulting cluster labels back to the *original* (unscaled) DataFrame.
  o To understand what each segment represents, calculate the mean of the original features for each cluster.
5. **Return Value**: The function should return a dictionary containing three key results:
  o `'optimal_pca_components'`: The number of components needed to explain 95% of the variance.
  o `'silhouette_score'`: The silhouette score of the final clustering.
  o `'cluster_profiles'`: A Pandas DataFrame containing the mean feature values for each customer segment, which helps in interpreting the clusters.
1.

Expt./Job No.:**6. Data Visualization Principles and Project**	Category:**Demonstration**

Learning Outcomes:- At the end of the experiment/job the student should be able to:

Sl No.	**Topic Learning Outcomes**	**COs**	**BL**
1	Compare the effectiveness of different visualization types and design principles to select the most appropriate chart for communicating a specific insight.	CO4	L4
2	Derive a compelling narrative from a complex dataset by analyzing its key features and creating a series of effective data visualizations to report the results.	CO4	L4

<u>Lesson Schedule:-</u>

Lecture No. - Portion covered per hour
1. Choosing the right chart for the data and message.
2. L - Creating a variety of charts (line, pie, area) and discussing their appropriate use cases.
3. Principles of clarity, conciseness, and visual hierarchy.
4. Project Update 1
5. The narrative structure of data stories.
6. Project Update 2
7. Best practices for designing effective data presentations.

8. Project Update 3
9. Misleading visuals and responsible data representation.
10. Final Project Demo
11. Open Discussion on EDA
12. Course Wrap-up

Review/Assignment Questions:-

Sl.No. - Questions

1. Project marks will be scale down to 20 marks.
 	
Project : Predictive Analytics for Agricultural Water Management in the Ujani Dam Catchment Area
 	
Area: Agriculture & Environmental Science
Location Focus: Solapur District, Maharashtra
 	
Problem Statement:
The Ujani Dam on the Bhima River is the primary water source for agriculture in the Solapur district. Farmers' livelihoods depend on the timely and adequate release of water, which is dictated by monsoon performance and dam storage levels. Your task is to build a model that analyzes historical data to provide actionable insights for local agricultural planning. You need to identify the key drivers of water availability and their impact on cropping patterns.
 	
Your Mission:
 	
1. **Data Acquisition:** Gather historical data from multiple sources:
  o Daily/weekly water storage levels and outflow from the Ujani Dam (from the Maharashtra Water Resources Department or similar public records).
  o Historical rainfall, temperature, and humidity data for the Solapur region from the Indian Meteorological Department (IMD).
  o (Optional) Use satellite imagery APIs (like Google Earth Engine) to estimate the green cover or cropped area in the region over the years.
2. **Data Analysis & Feature Engineering:**
  o Clean and merge the datasets. Handle missing values and time-series alignment.
  o Engineer features like "days since last significant rainfall," "rate of change in dam level," and "seasonal water surplus/deficit."
3. **Modeling & Visualization:**
  o Use correlation matrices and scatter plots to analyze the relationship between rainfall patterns and dam storage levels.
  o Develop a time-series visualization dashboard showing the interplay between rainfall, dam levels, and water outflow over the past decade.
  o **Challenge:** Apply Principal Component Analysis (PCA) on all weather variables to create a "Drought Index" component that summarizes adverse weather conditions. Analyze how this index correlates with low water levels.
 	
Final Deliverable: A detailed report with visualizations that helps answer: "Which climatic factors are the most reliable predictors of water stress in the Ujani catchment area?" and "Based on historical data, can we identify early warning signs of a low-water year?"

Project : Analyzing Customer Sentiment and Quality Control for the Solapur Textile Industry on E-commerce Platforms

Area: E-commerce & Natural Language Processing (NLP)
Location Focus: Solapur's Textile Industry

Problem Statement:
Solapur is famous for its textiles, especially the "Solapur Chaddar" (bedspreads) and towels, which are now sold on major e-commerce platforms across India. For small-scale manufacturers to thrive, they need to understand customer feedback to improve product quality and marketing. Your goal is to scrape and analyze thousands of customer reviews to provide actionable insights.

Your Mission:

1. **Web Scraping:** Build a web scraper using BeautifulSoup or Scrapy to collect product reviews for "Solapur Chaddar" or "Solapur Towel" from websites like Amazon.in or Flipkart.com. Extract the rating, the review text, and date.
2. **Text Cleaning and Preprocessing:**
   - Use regular expressions and string manipulation to clean the review text (remove punctuation, convert to lowercase).
3. **Sentiment Analysis and Topic Modeling:**
   - Apply a pre-trained sentiment analysis model to classify each review as positive, negative, or neutral.
   - Analyze the text of negative reviews. What are the common complaints? Use word clouds and frequency charts to identify keywords like "color fade," "shrink," "rough," "thin material."
   - **Challenge:** Use K-Means clustering on word frequency vectors (TF-IDF) of the negative reviews to automatically discover the main complaint categories (e.g., Cluster 1: "Poor Durability," Cluster 2: "Color/Dye Issues," Cluster 3: "Bad Customer Service").
4. **Visualization:**
   - Create bar charts showing the distribution of ratings.
   - Develop a dashboard presenting the key negative feedback themes and their frequency.

Final Deliverable: A business intelligence report for a Solapur textile association that highlights the top 3 areas for quality improvement and suggests marketing keywords based on what customers praise in positive reviews.

Project
Evaluation Rubric
(Total: 15 Marks)

This rubric assesses the project across four key areas: Data Management, Exploratory Analysis, Advanced Modeling, and Insight Generation.

Criterion	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (0-1 Mark)	Marks
1. Data Acquisition & Cleaning	All required datasets are successfully acquired, including complex sources (API/scraping). Data is meticulously cleaned: duplicates removed, missing values handled with a justified strategy (e.g., imputation, dropping), and data types are corrected. Code is efficient and well-documented.	Most data is acquired, but may struggle with one complex source. Standard cleaning procedures are applied correctly. Some minor inconsistencies may remain in the data.	Only the simplest datasets are acquired. Cleaning is superficial; for instance, missing values are dropped without considering the impact, or duplicates are missed.	Fails to load or properly merge critical data. The dataset remains messy and unsuitable for analysis. (0-1)	/ 4
2. Exploratory Data Analysis (EDA) & Visualization	A comprehensive suite of relevant visualizations (univariate, bivariate, and multivariate) is used. Plots are aesthetically pleasing, clearly labeled (titles, axes), and each one is accompanied by a concise, insightful observation that contributes to the overall narrative.	Several appropriate plots are created to explore the data. They are technically correct but may lack professional polish or insightful commentary. The exploration feels more like a checklist than a guided inquiry.	Basic plots are generated, but some may be inappropriate for the data type or poorly labeled. The EDA is superficial and does not uncover any significant patterns in the data.	Very few or incorrect visualizations are used. There is no clear evidence of data exploration. (0-1)	/ 4
3. Advanced Analysis & Modeling	The required advanced technique (PCA/K-Means/NLP) is implemented correctly, including all necessary preprocessing steps (e.g., standardization). There is a clear understanding of the model's parameters and evaluation (e.g., analyzing explained variance for PCA, using silhouette score for K-Means).	The advanced technique is implemented, but with minor conceptual or technical errors (e.g., applying K-Means to unscaled data). The student applies the model but may not fully justify their choices or evaluate the results robustly.	An attempt is made to apply the advanced technique, but the implementation has significant flaws. The results are likely incorrect or misleading due to these errors.	Fails to implement the required advanced technique or fundamentally misuses it. (0-1)	/ 4
4. Interpretation, Conclusion & Deliverable Quality		The final report/notebook provides a clear and compelling narrative. Results are interpreted deeply in the context of the problem (e.g., what a "cluster" means for Solapur farmers). The conclusion directly answers the project statement with actionable recommendations strongly supported by the analysis. The entire deliverable is professional and well-structured.	A conclusion is drawn, but it may be generic or not fully supported by all aspects of the analysis. The interpretation of results is present but lacks depth. The final report is complete but could be better organized.	No meaningful interpretation or conclusion is offered. The deliverable is just a dump of code and outputs without context or a story. (0-1)	/ 3

Self & Peer Evaluation for EDA Project

Instructions: The goal of this evaluation is to reflect on the group's dynamics and individual contributions to the project. Please provide an honest and constructive assessment of yourself and each of your group members based on their performance throughout the project. Rate each person on a scale of 1 to 4 for the 6 categories below. Total the score for each individual.

Group Member Names:

1. **My Name:** _____
2. **Member's Name:** _____
3. **Member's Name:** _____
4. **Member's Name:** _____
5. **Member's Name:** _____

Evaluation Category	1. (Me)	2.	3.	4.	5.
Collaboration					
Active Participation					
Task Completion & Preparation					
Quality of Code & Analysis					
Technical Skill & Application					
Communication					
TOTAL SCORE (out of 24)					

2. Active Participation
The way in which this person participated in work sessions was:

- **4 -** An active driver. Constantly proposed analytical steps, wrote code, generated plots, and actively sought out new ways to explore the data.
- **3 -** Contributed when asked. Wrote code and performed analysis but primarily followed tasks assigned by others rather than proposing new directions.
- **2 -** Was mainly passive. Observed others coding and analyzing but contributed little original work.
- **1 -** Was mentally or physically absent. Did not contribute any effort during collaborative work time.

3. Task Completion & Preparation
Regarding individually assigned tasks, this person:

- **4 -** Consistently completed their assigned tasks (e.g., cleaning a column, creating a set of plots) ahead of schedule and often did more than required, such as helping others.
- **3 -** Reliably completed the tasks they committed to doing on time.
- **2 -** Completed some, but not all, of their assigned tasks. Their work was often late or incomplete, holding the group back.
- **1 -** Did little or none of the work they were assigned.

4. Quality of Code & Analysis
The typical quality of the work this person submitted to the group was:

- **4 -** Work was of high quality. Code was clean, efficient, and well-commented. Analysis and visualizations were insightful, accurate, and directly addressed project goals.
- **3 -** Work was adequate. Code functioned correctly but might lack comments or efficiency. Analysis was generally accurate but sometimes superficial.
- **2 -** Work needed significant revision. Code was often buggy, inefficient, or difficult for others to understand. Analysis contained some errors or misinterpretations.
- **1 -** Work was unusable. The code did not run or the analysis was fundamentally flawed, hindering the group's progress.

5. Technical Skill & Application
This person displayed technical knowledge of data analysis (e.g., Pandas, Matplotlib, Stats) such as:

- **4 -** Demonstrated a strong understanding of the course's technical concepts and could apply them independently and correctly to the project's data.
- **3 -** Had a good general grasp of the material but occasionally needed help from others with more complex functions or statistical concepts.
- **2 -** Frequently struggled with the core technical skills required for the project, requiring significant help from others to complete their tasks.
- **1 -** Relied almost entirely on others for all technical work. Showed little understanding of the code or analysis being performed.

6. Communication
The way in which this person communicated was:

- **4 -** Highly reliable and proactive. Regularly updated the group on progress and roadblocks. Was easily reachable and frequently checked group messages/email.
- **3 -** Was a fairly good communicator. Stayed in touch but was occasionally difficult to contact or slow to respond.
- **2 -** Communication was infrequent. Was often unreachable and rarely provided updates on their work without being prompted.
- **1 -** Was unreachable. Did not communicate with group members about the project, their progress, or their availability.

CIA Rubrics

Assessment Occasion: Data_Manipulation_with_Pandas

Sl No.	**Criteria**	**CO Code**	**Scale of Assessment**		
1	Data Loading & Initial Inspection	CO2	**0-1** Need Improvment	**2** Proficient	**3** Excellent
			Loads the data, but inspection is minimal, used incorrectly, or	Correctly loads data and performs basic inspection	Correctly loads data and uses a comprehensive combination of functions (.head(), .info(),

			missing key initial checks.	(.head(), .info()), but misses a more thorough statistical or structural check (e.g., no .describe()).	.describe(), .shape) to provide a full overview of the dataset's structure, types, and summary statistics.
2	Data Cleaning & Preprocessing	CO2	**0-2** Needs Improvment	**3-4** Proficient	**5** Excellent
			Attempts cleaning, but the approach is incomplete or flawed (e.g., identifies but doesn't handle issues, or drops excessive data without justification). No significant or functional data cleaning is performed.	Identifies and handles most major issues (e.g., missing values) but may miss subtle ones (e.g., data types) or provides weak justification for the cleaning methods used.	Systematically identifies and handles missing values, duplicates, and incorrect data types using appropriate methods (fillna, dropna, astype, drop_duplicates). Provides clear justification for all cleaning choices.
3	Exploratory Analysis & Insights	CO3	**0-2** Need Improvment	**3-4** Proficient	**5** Excellent
			Calculates basic statistics (.mean(), .value_counts()) but without	Uses aggregation to produce summary	Effectively uses aggregation methods (.groupby(),

			meaningful grouping or comparison. Findings are stated without context or interpretation.	statistics. The analysis is present but mostly descriptive (the "what") rather than interpretive (the "so what" or insight).	.pivot_table()) to answer 2-3 meaningful analytical questions. Provides clear, well-written interpretations of the results that reveal interesting patterns or insights.
4	Code Quality & Documentation	CO3	**0-1** Need Improvment	**2** Good	-
			Code is functional but could be better organized. Documentation or good naming conventions are used inconsistently.	Code is clean, well-structured, and easy to follow. Uses meaningful variable names and is thoroughly documented with comments and Markdown to explain the logic.	-

Assessment Occasion: Data_Collection_and_Cleaning

Sl No.	Criteria	CO Code	Scale of Assessment		
1	Data Collection / Acquisition	CO2	**0-2** Developing	**3** Proficient	**4** Excellent
			An attempt to collect or load data is made, but it is incomplete, contains errors, or requires significant manual correction to work.	Successfully collects or loads the primary dataset. The process is functional but may lack efficiency or clear documentation. The chosen method is adequate but not thoroughly justified.	Successfully collects data from a suitable source (e.g., API, web scrape, multiple files). The collection method is robust, efficient, and well-documented. If loading a file, necessary parameters (e.g., encoding, delimiter) are used correctly.
2	Initial Data Inspection & Problem Identification	CO3	**0-2** Developing	**3** Proficient	**4** Excellent
			Performs only minimal inspection (e.g., just .head()). The identification of data quality problems is	Conducts standard inspection (.info(), .head()) and identifies the most obvious	Performs a thorough inspection using a variety of functions (.info(), .describe(),

			vague, incomplete, or absent.	problems (e.g., missing values). May overlook more subtle issues (e.g., inconsistent categories) or fails to provide a concise summary.	.isnull().sum(), .value_counts(), .shape). Clearly articulates and summarizes all major and subtle data quality issues found.
3	Handling Missing Data & Duplicates	CO2	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Attempts to handle either missing data or duplicates, but not both. The method is applied incorrectly or is clearly inappropriate for the context (e.g., fillna(0) on a column where 0 is a meaningful value).	Identifies and handles both missing values and duplicates, but the method used may be suboptimal (e.g., dropping too much data) or the justification is weak or missing.	Systematically identifies and handles both missing values and duplicate records. The chosen strategy (e.g., imputation, dropping rows/columns) is highly appropriate for the data and is explicitly justified with clear reasoning.
4	Data Formatting & Standardization	CO2	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Attempts to fix a single data type or format issue,	Corrects the most critical data type	Thoroughly corrects data types (astype,

			but the approach is incomplete, flawed, or applied to a trivial problem.	issues and performs some basic standardization (e.g., fixing column names). Overlooks more nuanced inconsistencies within the data itself (e.g., 'NY' vs 'New York').	to_numeric), standardizes text data (e.g., strips whitespace, unifies case), fixes structural inconsistencies (e.g., consolidates categories), and ensures column names are clean and consistent.
5	Code Quality & Reproducibility	CO2	**0-1** Needs Improvment	**2** Good	-
			Code is functional but could be better structured. Documentation is present but inconsistent or lacks detail, making some of the cleaning logic difficult to follow.	clean, efficient, and well-organized. Markdown cells and comments clearly explain each step of the cleaning process and the reasoning behind decisions. The entire notebook/script is easily reproducible from start to finish.	-

Assessment Occasion: Advanced_EDA_and_Dimensionality_Reduction

Sl No.	Criteria	CO Code	Scale of Assessment		
1	Multivariate & Bivariate Analysis	CO4	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Creates basic bivariate plots (scatter, bar) but fails to explore multivariate relationships. Plot choices may not be the most effective for the data, or labels/titles are missing.	Creates several appropriate visualizations, including at least one multivariate plot (like a heatmap). Plots are functional and clear but may lack the variety or polish of an excellent submission.	Creates a rich variety of advanced visualizations (e.g., correlation heatmap, pair plots, violin plots, faceted plots). Plots are well-labeled, aesthetically clean, and perfectly chosen to investigate specific relationships between variables.
2	Insight Generation & Hypothesis Formulation	CO3	**0-2** Developing	**3** Proficient	**4** Excellent
			States only obvious facts from plots without providing deeper interpretation or context (e.g., "The	Accurately describes the patterns observed in the plots (e.g., "X and Y have	Provides deep, well-articulated interpretations of the EDA findings. Connects

			bar for Group A is higher than Group B").	a strong positive correlation"). The insights are correct but may be superficial, failing to connect different findings or suggest next steps.	insights from multiple plots to form a cohesive narrative or data story. Formulates clear, data-driven hypotheses for further investigation.
3	Preprocessing for Dimensionality Reduction	CO3	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Applies PCA directly to unscaled data without acknowledging the critical need for scaling. The technical implementation of PCA itself is mostly correct.	Attempts to apply scaling but the implementation has minor flaws (e.g., scales the entire dataset before splitting). Alternatively, applies PCA without scaling but explicitly notes this limitation and its potential impact.	Correctly identifies and articulates the need for feature scaling (e.g., StandardScaler) before applying PCA. Implements the scaling process flawlessly, ensuring the model is fit on training data only (if applicable) and transformed correctly.
4	Application & Interpretation of PCA	CO3	**0-2** Developing	**3** Proficient	**4** Excellent

			Runs PCA but does not analyze the results. Fails to determine an appropriate number of components or visualize the transformed data.	Applies PCA and creates an explained variance plot but makes a weak or arbitrary choice for the number of components. The visualization of the transformed data is present but lacks meaningful interpretation.	Correctly applies PCA and uses the cumulative explained variance plot (or a similar method) to make a well-justified decision on the number of components to retain. Visualizes the data in the new principal component space (e.g., 2D scatter plot) and interprets the resulting patterns.
5	Code Quality & Documentation	CO4	**0-1** Needs Improvment	**2** Good	-
			The code is functional, but the notebook's structure or documentation is inconsistent. Some parts are well-explained, while others lack comments or clear organization.	The entire notebook is exceptionally well-structured, with clear Markdown headings, comments, and text cells that narrate the analytical process. The code is clean,	-

				efficient, and easily reproducible.	

Assessment Occasion: Project

Sl No	Criteria	CO Code	Scale of Assessment		
1	Problem Definition & Data Sourcing	CO3	**0-1** Developing	**2** Proficient	**3** Excellent
			The problem statement is vague or overly simplistic. The dataset is chosen without clear justification, or the acquisition is incomplete.	States a clear problem. The chosen dataset is appropriate, but the connection between the problem and the data could be stronger. The acquisition process is functional.	Clearly articulates a well-defined problem or question. Selects and acquires a relevant and sufficiently complex dataset. Provides strong justification for why the problem is interesting and the data is suitable.
2	Data Cleaning & Feature Engineering	CO3	**0-2** Developing	**3-4** Proficient	**5** Excellent

			Cleaning is incomplete; some obvious issues are missed. Only basic cleaning is performed (e.g., dropping all NaNs without consideration). No feature engineering is attempted.	Performs all necessary cleaning steps correctly. Feature engineering is attempted and functional, but the new features may be simple or their value is not fully demonstrated.	Executes a thorough and systematic cleaning process. Thoughtfully handles missing data, outliers, and inconsistencie s with clear justification. Creatively engineers 2+ new features that are meaningful and add value to the analysis or model.
3	Exploratory Data Analysis (EDA) & Visualization	CO3,CO 4	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Creates a limited number of basic plots (e.g., histograms, simple bar charts). The analysis is superficial, or plots are poorly chosen or labeled.	Produces a good range of correct and relevant visualizations. The analysis is descriptive and accurate, but may lack the depth or narrative flow of an excellent submission.	Conducts a comprehensiv e EDA with a strategic mix of univariate, bivariate, and multivariate visualizations. Plots are insightful, well-labeled, and effectively used to uncover patterns and

					test initial hypotheses. The EDA tells a clear story about the data.
4	Analysis, Modeling, & Interpretation	CO4	**0-2** Developing	**3-4** Proficient	**5** Excellent
			Analysis is a simple restatement of facts from the EDA. If modeling, the implementation is flawed (e.g., no train/test split) or the results are not interpreted.	Derives valid conclusions from the EDA. If modeling, the model is implemented correctly, but the evaluation or interpretation is basic. The link back to the original problem is present but not deeply explored.	Synthesizes EDA findings into powerful insights. If modeling, selects an appropriate model, implements it correctly (including train/test split), and evaluates its performance meaningfully. Provides a deep interpretation of results in the context of the original problem.
5	Conclusion & Communication	CO4	**0-1** Need Improvment	**2** Good	-
			The project has a conclusion, but	Delivers a well-structured and professional-	-

			it may be brief or simply repeat earlier points. The notebook is functional but could be better organized or documented.	quality report/notebook . Provides a concise conclusion summarizing key findings, discusses limitations, and suggests future work. Code is clean, well-commented, and easily reproducible.	

TEE Model Question Paper

Question Paper Title: End Sem Exam		
Total Duration (H:M): 3:0	**Course: Exploratory Data Analysis and Data Visualization (SC2301)**	**Maximum Marks: 20**

1	**End-of-Semester Examination**
	
	Question : Real Estate Data Analysis (Total: 15 Marks)
	
	You are a junior data analyst at a real estate firm. You have been given a small dataset containing information about recent property sales. Your task is to clean the data, perform an exploratory analysis, and present your key findings.
	
	Given Dataset:
	
	`import pandas as pd`
	`import numpy as np`
	
	`data = {`
	`'property_id': [101, 102, 103, 104, 105, 106, 107, 108],`
	`'location': ['Suburb', 'Downtown', 'Suburb', 'Rural', 'downtown', 'Suburb', 'Rural', 'Suburb'],`
	`'num_bedrooms': [3, 4, 3, 5, 4, 2, 4, 3],`
	`'num_bathrooms': [2.0, 3.0, 2.0, 4.0, 3.0, 1.0, None, 2.0],`
	`'square_feet': [1800, 2200, 1950, 3500, 2400, 1200, 3100, 1750],`
	`'price': [350000, 650000, 380000, 750000, 680000, 250000, 695000, 345000]`
	`}`
	
	Instructions:
	
	Write Python code using the Pandas library to perform the following tasks. Add comments or markdown cells to explain your steps where necessary.
	
	Part A: Data Loading and Cleaning (4 Marks)
	
	1. Create a Pandas DataFrame from the `data` dictionary provided.
	2. Identify and handle the missing value in the `num_bathrooms` column by filling it with the **median** value of that column. Justify why the median is an appropriate choice here.
	3. Standardize the `location` column so that categories like 'Downtown' and 'downtown' are treated as the same category ('Downtown').
	
	Part B: Feature Engineering (2 Marks)
	
	1. Create a new column in your DataFrame called `price_per_sq_ft` which is calculated by dividing the `price` by the `square_feet`. This will help in comparing property values more effectively.
	
	Part C: Exploratory Data Analysis & Interpretation (6 Marks)
	
	1. Calculate and display the **average price** for properties in each `location`.
	
	2. Calculate and display the **correlation matrix** for the numerical columns: `price`, `square_feet`, and `num_bedrooms`.
	3. Based on your findings in C1 and C2, write a brief summary (2-3 sentences) of your key insights. For instance, which location is the most expensive on average? Which features have the strongest relationship with the property price?
	
	Part D: Data Visualization (3 Marks)
	
	1. Create a well-labeled **bar chart** to visualize the average property price for each location (as calculated in Part C1).
	2. Ensure your chart includes a title, an x-axis label, and a y-axis label.
		15.00	CO1,CO2,CO3,CO4	L2, L3, L4

2	1. Create a well-labeled **bar chart** to visualize the average property price for each location (as calculated in Part C1). 2. Ensure your chart includes a title, an x-axis label, and a y-axis label. 1. What is the key difference between a Python `list` and a `tuple`? Can you give a practical example of when you would choose to use a tuple over a list in a data analysis context? 2. Can you explain what a Pandas DataFrame is and how it differs from a 2D NumPy array? What specific advantages does a DataFrame offer for data cleaning and exploration? 3. Imagine you load a dataset and discover a column has 20% missing values. What are your primary options for handling this, and what is the key factor you would consider when deciding whether to **impute** the values versus **dropping** the rows or the entire column? 4. When analyzing a feature like "salary" or "house price," why is the **median** often considered a more reliable measure of central tendency than the **mean**? What kind of plot would you use to visually confirm the distribution of such a feature? 5. In any machine learning project, why is it considered essential to split your data into a training set and a testing set? What critical problem are you trying to evaluate and prevent by doing this?	5.00	CO1,CO2,CO3,CO4	L2, L3, L4

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Course Plan

Semester: 01 Year: 2024-25

Course Title: Introduction to Data Science	Course Code: SC2303
L-T-P : 3-0-0	Credits: 03
Total Contact hrs: 45	Duration of ESA: 3 hours
ISA Weightage: 50	ESA Weightage:50
Lesson Plan Authors: Dr. Savita S. Hanji	Date:

At the end of the course the student will be able to:
CO 1. Describe the data science terminologies
CO 2. Apply the Data Science process on real time scenario.
CO 3. Analyze data visualization tools
CO 4. Apply Data storage and processing with frameworks

UNIT- I

PREPARING AND GATHERING DATA AND KNOWLEDGE (9 Hours)

Philosophies of data science – Data science in a big data world – Benefits and uses of data science and big data – facts of data: Structured data, Unstructured data, Natural Language, Machine generated data, Audio, Image and video streaming data – The Big data Eco system: Distributed file system, Distributed Programming framework, Data Integration frame work, Machine learning Framework, NoSQL Databases, Scheduling tools, Benchmarking Tools, System Deployment, Service programming and Security

THE DATA SCIENCE PROCESS (10 Hours)

Overview of the data science process- defining research goals and creating project charter, retrieving data, cleansing, integrating and transforming Data, Statistics for data science, exploratory data analysis.

UNIT -II

MACHINE LEARNING (16 Hours)

Application for machine learning in data science- Tools used in machine learning- Modeling Process – Training model – Validating model – Predicting new observations –Types of machine learning Algorithm : Supervised learning algorithms, Unsupervised learning algorithms. Build the models, presenting findings and building application on top of them.

UNIT- III

VISUALIZATION (5 Hours)

Introduction to data visualization – Data visualization options – Filters – MapReduce – Dashboard development tools.

CASE STUDIES (5 Hours)

Distributing data storage and processing with frameworks – Case study: e.g, Assessing risk when lending money.

Textbooks

1. Introducing Data Science, Davy Cielen, Arno D. B. Meysman and Mohamed Ali,Manning Publications, 2016.

Reference Books

1. Doing Data Science, Straight Talk from the Frontline, Cathy O'Neil, Rachel Schutt, O' Reilly, 1st edition, 2013.
2. Mining of Massive Datasets, Jure Leskovec, Anand Rajaraman, Jeffrey David Ullman, Cambridge University Press, 2nd edition, 2014
3. An Introduction to Statistical Learning: with Applications in R, Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Springer, 1st edition, 2013
4. Think Like a Data Scientist, Brian Godsey, Manning Publications, 2017.

ISA Evaluation Scheme*

Assessment	Weightage in Marks
Minor 1	20
Minor 2	20
Assignment	10
Total	50

SCBTCSE24-25/Rev.1.0

Course Plan

Semester: 01 Year: 2024-25

Course Title: Business Communications	Course Code: BS-2006
L-T-P : 0-0-3	Credits: 03
Total Contact hrs: 90	Duration of ESA: 3 hours
ISA Weightage: 50	ESA Weightage:50

Syllabus:

- Digital Communication Etiquette
- Written Messages: Neutral & Negative Messages
- Written Messages: Customer Comments
- In-Person Communication: Spoken English & Pronunciation
- In-Person Communication: Common Errors in English
- In-Person Communication: Active Listening and Empathy
- In-Person Communication: Mock Networking & Elevator Pitch
- In-Person Communication: Public Speaking
- In-Person Communication: Handling Q&A Sessions
- Preparing for a job interview
- Managing your online image & Practicing business etiquette
- Business Writing: Memos, Reports, Proposals
- Business Writing: Agendas and Minutes of meetings
- Employment Communication: Resume Writing
- Employment Communication: Writing Cover Letters
- Employment Communication: Writing Inquiry Emails
- Sensitivity to Cultural and Gender-Related Issues
- Presentation Skills: Planning and organizing a presentation
- Presentation Skills: Visual Aids and digital tools

ISA Evaluation Scheme*

Assessment	Weightage in Marks
Evaluation 1	10*
Evaluation 2	20*
Evaluation 3	20*
Evaluation 4	20*
Evaluation 5	10
Total	50